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Code No. 1J095-8916-8

D1005-E4 · D1105-E4 · D1305-E4 · V1505-E4 · D1105-K-E4

OPERATOR'S MANUAL KUBOTA DIESEL ENGINE

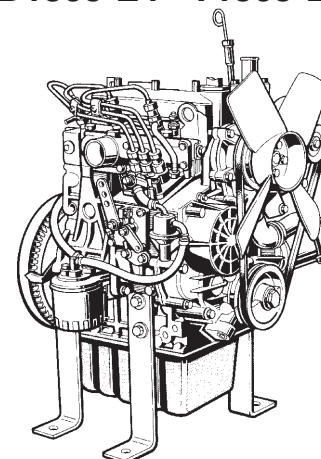
MANUEL DE L'OPERATEUR MOTEUR DIESEL DE KUBOTA

BEDIENUNGSANLEITUNG KUBOTA DIESEL MOTOR

MANUALE DELL'OPERATORE KUBOTA MOTORE DIESEL

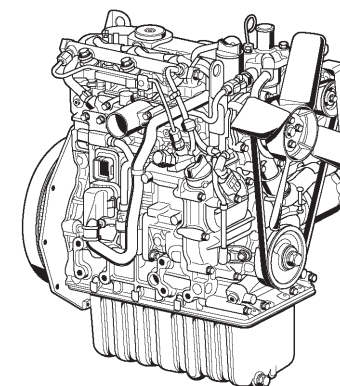
MANUAL DEL OPERADOR MOTOR DIESEL KUBOTA

D1005-E4 · D1105-E4
D1305-E4 · V1505-E4



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D1105-K-E4



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READ AND SAVE THIS BOOK
MANUEL A LIRE ET A CONSERVER
DIESE ANLEITUNG SORGFÄLTIG DURCHLESEN
UND GRIFFBEREIT AUFBEWAHREN
LEGGETE E CONSERVATE QUESTO MANUALE
LEAN Y CONSERVEN ESTE MANUAL

California Proposition 65

⚠ WARNING ⚠

Engine exhaust, some of its constituents,
certain vehicle components and fluids,
contain or emit chemicals known to the
State of California to cause cancer and birth
defects or other reproductive harm.

OPERATOR'S MANUAL

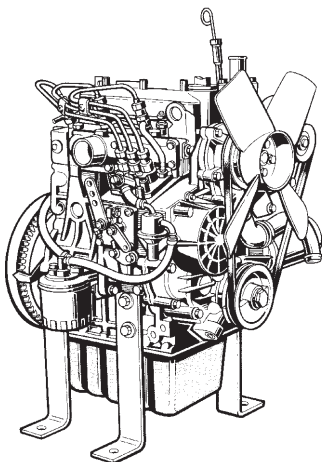
ENGLISH

KUBOTA DIESEL ENGINE

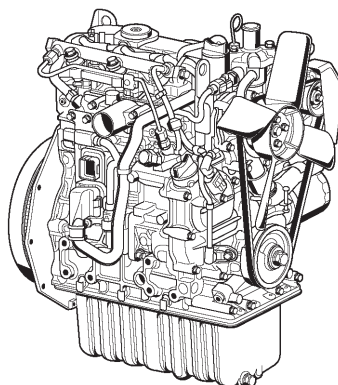
MODELS

**D1005-E4 · D1105-E4
D1305-E4 · V1505-E4**

D1105-K-E4



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READ AND SAVE THIS MANUAL

Kubota

FOREWORD

You are now the proud owner of a KUBOTA Engine. This engine is a product of KUBOTA quality engineering and manufacturing. It is made of fine materials and under a rigid quality control system. It will give you long, satisfactory service. To obtain the best use of your engine, please read this manual carefully. It will help you become familiar with the operation of the engine and contains many helpful hints about engine maintenance. It is KUBOTA's policy to utilize as quickly as possible every advance in our research. The immediate use of new techniques in the manufacture of products may cause some small parts of this manual to be outdated. KUBOTA distributors and dealers will have the most up-to-date information. Please do not hesitate to consult with them.



SAFETY FIRST

This symbol, the industry's "Safety Alert Symbol", is used throughout this manual and on labels on the machine itself to warn of the possibility of personal injury. Read these instructions carefully. It is essential that you read the instructions and safety regulations before you attempt to assemble or use this unit.



DANGER : Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.



WARNING : Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION : Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.

IMPORTANT : Indicates that equipment or property damage could result if instructions are not followed.

NOTE : Gives helpful information.

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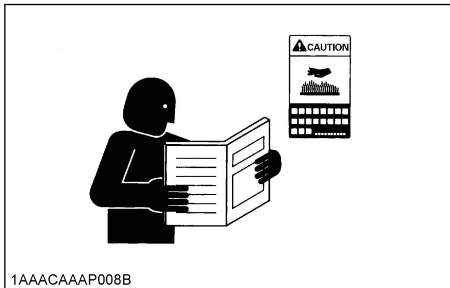
SAFE OPERATION

Careful operation is your best assurance against an accident. Read and understand this section carefully before operating the engine. All operators, no matter how much experience they may have, should read this and other related manuals before operating the engine or any equipment attached to it. It is the owner's obligation to provide all operators with this information and instruct them on safe operation.

Observe the following for safe operation.

OBSERVE THE SAFETY INSTRUCTIONS

- Carefully read and understand this operator's manual and safety labels on the engine before attempting to start and operate the engine.



- Learn how to operate and work safely. Know your equipment and its limitations. Always keep the engine in good condition.
- Before allowing other people to use your engine, explain how to operate it, and have them read this manual before operation.
- Do not modify the engine. Unauthorized modifications may impair the engine's function and/or safety and affect the lifespan of the engine. If the engine does not perform properly, consult your local Kubota Engine Distributor first.

WEAR SAFE CLOTHING AND PERSONAL PROTECTIVE EQUIPMENT (PPE)

- Do not wear loose, torn, or bulky clothing when working around the machine. Such clothing may get caught on working controls and projections or in fans, pulleys, and other moving parts and cause personal injury.
- Use additional safety items and PPE such as a hard hat, safety protection, safety goggles, and gloves as appropriate or as required.



- Do not operate the machine or any equipment attached to it while under the influence of alcohol, medication, or other drugs, or while fatigued.
- Do not wear radio or music headphones while operating the engine.

CHECK BEFORE STARTING AND OPERATING THE ENGINE

- Inspect the engine before operation. Do not operate the engine if there is something wrong with it. Repair it immediately.

SAFE OPERATION



1BAABADAP0010

- Ensure all guards and shields are in place before operating the engine. Replace any guards and shields that are damaged or missing.
- Check to see that you and others are a safe distance from the engine before starting.
- Always keep the engine at least 3 ft (1 m) away from buildings and other facilities.
- Do not allow children or livestock to approach the machine while the engine is running.
- Do not start the engine by shorting across starter terminals. The machine may start in gear and move unexpectedly. Do not bypass or defeat any safety devices.

KEEPING THE ENGINE AND SURROUNDINGS CLEAN

- Stop the engine before cleaning.
- To prevent fires, keep the engine clean and free of accumulated dirt, grease, and trash. Store flammable fluids in proper containers and cabinets away from sparks and heat.



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- Check for leaks and repair any leaks immediately.
- Do not stop the engine without idling. Allow the engine to cool down first. Keep the engine idling for about 5 minutes before stopping unless there is a safety problem that requires immediate shut down.

SAFE HANDLING OF FUEL AND LUBRICANTS - KEEP AWAY FROM FIRE

- Always stop the engine before refueling and/or lubricating.
- Do not smoke or allow flames or sparks in the work area. Fuel is extremely flammable and explosive under certain conditions.



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- Refuel in a well ventilated and open place. If fuel and/or lubricant are spilled, allow the engine to cool down before refueling.
- Do not mix gasoline or alcohol with diesel fuel. The mixture can cause a fire or severe engine damage.
- Do not use unapproved containers such as buckets, bottles, and jars. Use approved fuel storage containers and dispensers.

EXHAUST GASES AND FIRE PREVENTION

- Engine exhaust fumes can be very harmful if allowed to accumulate. Run the engine in a well ventilated location and where there are no people or livestock near the engine.
- The exhaust gas from the muffler is very hot. To prevent fires, do not expose dry grass, mowed grass, oil or any other combustible materials to exhaust gas. Keep the engine and muffler clean at all times.



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- To prevent fires, be alert for leaks of flammable substances from hoses and lines. Check hoses and pipes for leaks of flammable substances, such as fuel and hydraulic fluid, by following the maintenance check list.
- To prevent fires, do not short across power cables and wires. Check to see that all power cables and wiring are in good condition. Keep all electrical connections clean. Exposed wire or frayed insulation can cause a dangerous electrical shock and personal injury.

ESCAPING FLUID

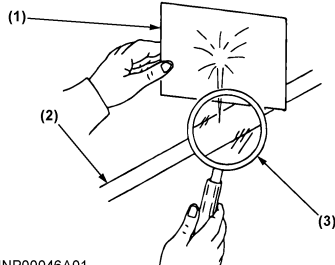
- Relieve all pressure in the air system, the oil system, and the cooling system before disconnecting any lines, fittings, or related items.
- Be cautious of possible pressure relief when disconnecting any device from a pressurized system that utilizes pressure. Do not check for pressure leaks with your hand. High pressure oil or fuel can cause personal injury.
- Escaping fluid under pressure has sufficient force to penetrate skin causing serious personal injury.



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- Fluid escaping from pinholes may be invisible. Use a piece of cardboard or wood

to search for suspected leaks. Do not use your hands or your body. Use safety goggles or other eye protection when checking for leaks.



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- (1) Cardboard
- (2) Line
- (3) Magnifying glass

- If injured by escaping fluid, see a medical doctor immediately. This fluid can produce gangrene or a severe allergic reaction.

CAUTIONS AGAINST BURNS AND BATTERY EXPLOSIONS

- To avoid burns, be cautious of hot components, such as the muffler, the muffler cover, the radiator, the hoses, the engine body, the coolants, and the engine oil, during operation and after the engine has been shut off.

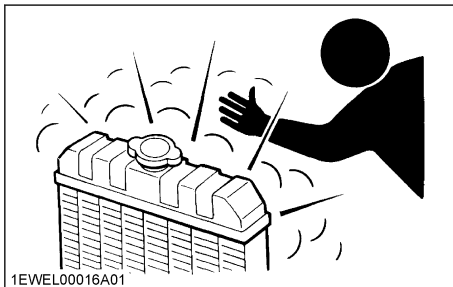


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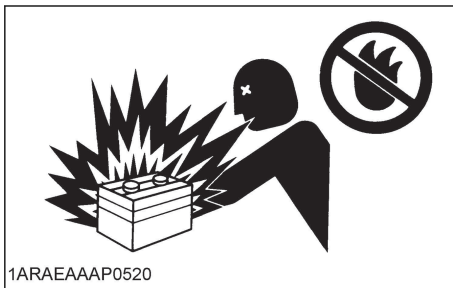
- Do not remove the radiator cap while the engine is running or immediately after stopping. Otherwise hot coolant will spout out from the radiator. Wait until the radiator is completely cool to the touch before removing the cap. Wear safety goggles.

SAFE OPERATION

ENGLISH

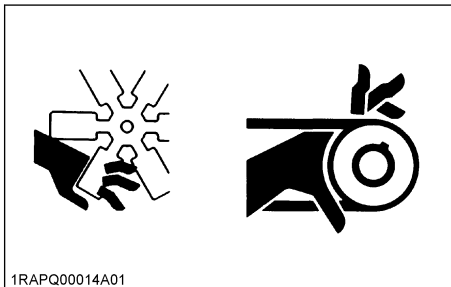


- Close the coolant drain plugs, secure the pressure cap, and fasten the pipe bands before operating. If the engine is operated when these parts are loose or have been removed, serious personal injury can occur.
- The battery presents an explosive hazard. When the battery is being charged, hydrogen and oxygen gases are extremely explosive.
- Do not use or charge the battery if its fluid level is below the **[LOWER]** mark. Otherwise, the component parts may deteriorate more quickly than expected, which may shorten the service life of the engine, or cause an explosion. Immediately add distilled water until the fluid level is between the **[UPPER]** and **[LOWER]** marks.
- Keep sparks and open flames away from the battery, especially during charging. Do not strike a match near the battery.
- Do not check the battery charge by placing a metal object across the terminals. Use a voltmeter or a hydrometer.
- Do not charge the battery when it is frozen. There is a risk of explosion. If the battery is frozen, warm it up to at least 16°C (61°F) before charging.



KEEP HANDS AND BODY AWAY FROM ROTATING PARTS

- Stop the engine before checking or adjusting the belt tension and cooling fan.
- Keep your hands and body away from rotating parts such as the cooling fan, the V-belt, the fan drive pulley and the flywheel. Contact with the rotating parts can cause severe personal injury.



- Do not run the engine without safety guards. Securely install the safety guards before operation.

HANDLING ANTIFREEZE

- Antifreeze is poisonous. If you swallow the antifreeze, seek immediate medical help. Do not make a person throw up unless told to do so by a poison control or a healthcare professional. Use standard first aid and CPR for signs of shock or cardiac arrest. Call your local poison control center or your local emergency number for further assistance.
- When handling antifreeze, use protective clothing such as rubber gloves.
- When antifreeze comes in contact with the skin or clothing, wash it off immediately.



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- Do not mix different types of antifreeze. The mixture can produce a chemical reaction resulting in harmful substances. Use genuine Kubota antifreeze or Kubota approved antifreeze.

DISPOSAL OF FLUIDS

- Be mindful of the environment and the ecosystem. Before draining any fluids, determine the correct way to dispose of them. Observe the relevant environmental protection regulations when disposing of oil, fuel, coolant, brake fluid, filters, or batteries.
- When draining fluids from the engine, place a suitable container underneath the engine body.
- Do not pour waste onto the ground, down a drain, or into any water source. Dispose of waste fluids according to environmental regulations.



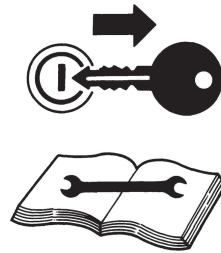
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CONDUCTING SAFETY CHECKS AND MAINTENANCE

- When inspecting or servicing the engine, place the engine on a large flat surface. Do not work on anything that is only supported by lift jacks or a hoist. Always use blocks or

the correct stands to support the engine when servicing.

- Disconnect the battery from the engine before conducting service. Put a "DO NOT OPERATE" tag on the starter switch to prevent the engine from being started accidentally.
- To avoid sparks from an accidental short circuit always disconnect the battery's ground cable (-) first and reconnect it last.
- Stop the engine and remove the key when conducting daily and periodic maintenance, service, or cleaning.



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- Only perform checks and maintenance after the engine, the coolant, the muffler, and the muffler cover have cooled off completely.
- Always use the appropriate tools and fixtures, and verify that they are in good condition before performing any service work. Make sure you understand how to use them before servicing the engine.
- Use only correct engine barring techniques for manually rotating the engine. Do not attempt to rotate the engine by pulling or prying on the cooling fan and V-belt. This practice can cause serious personal injury or premature damage to the cooling fan and belt.
- Replace the fuel pipes and the lubricant pipes along with their hose clamps every 2 years or sooner whether they are damaged or not. They are made of rubber and age gradually.
- When servicing is performed together by two or more persons, take care to perform all work safely.
- Keep a first aid kit and the appropriate type of fire extinguisher nearby at all times.

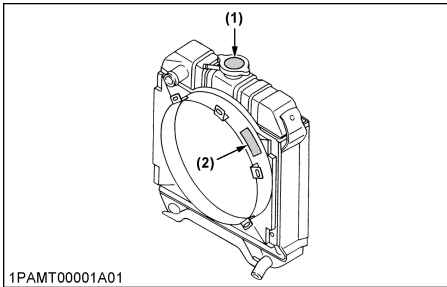


SAFE OPERATION

SAFETY LABELS

ENGLISH

1. Safety label locations



(1)

The following label is just an example. A different label might be applied on the actual machine.



(2) Part No. TA040-4957-1

Keep your hands away from the engine fan and the fan belt.



2. Care of safety labels

- Keep safety labels clean and free from obstructing material.
- Clean safety labels with soap and water, and dry with a soft cloth.
- Replace damaged or missing safety labels with new labels from your Kubota dealer.
- If a component with safety label(s) affixed is replaced with new part, make sure new label(s) is (are) attached in the same location(s) as the replaced component.
- Mount new safety labels by applying on a clean dry surface and pressing any bubbles to outside edge.

SERVICING THE ENGINE

Your dealer has knowledge of your new engine and has the desire to help you get the most value from it. After reading this manual thoroughly, you will find that you can do some of the regular maintenance yourself.

However, when in need of parts or major service, see your Kubota Dealer.

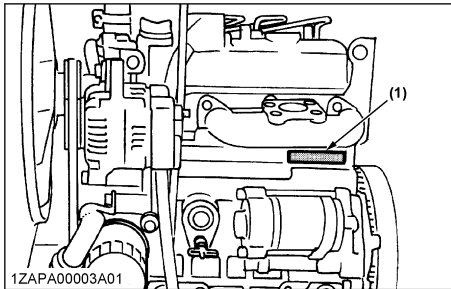
For service, contact the Kubota Dealership from which you purchased your engine or your local Kubota Dealer.

When in need of parts, be prepared to give your dealer the engine serial number.

Locate the serial number now and fill in the following tables.

Dealer's name (To be filled in by the owner)	
Date of purchase	

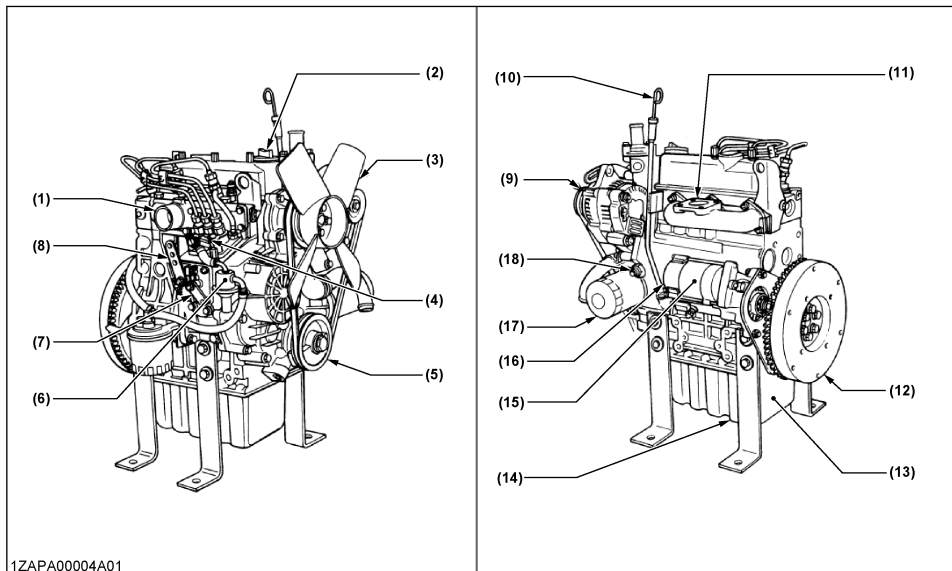
	Model	Serial number
Engine		



(1) Engine serial number

NAMES OF PARTS

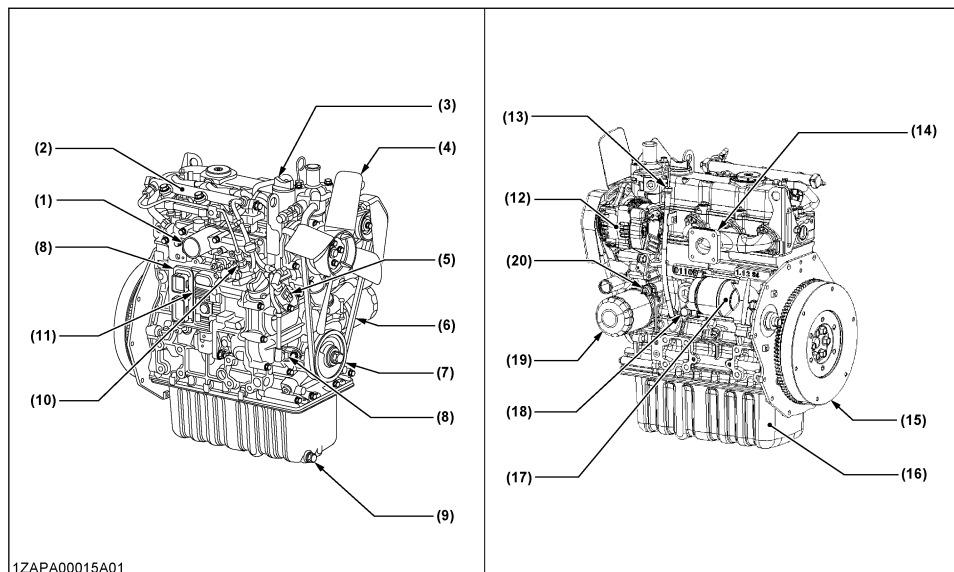
[D1005-E4, D1105-E4, D1305-E4, V1505-E4]



- (1) Intake manifold
- (2) Oil filler plug
- (3) Cooling fan
- (4) Injection pump
- (5) Fan drive pulley
- (6) Fuel feed pump
- (7) Engine stop lever
- (8) Speed control lever

- (9) Alternator
- (10) Oil level gauge
- (11) Exhaust manifold
- (12) Flywheel
- (13) Oil pan
- (14) Oil drain plug
- (15) Starter
- (16) Water drain plug
- (17) Oil filter cartridge
- (18) Oil pressure switch

[D1105-K-E4]



- (1) Intake manifold
- (2) Holder
- (3) Oil filler plug
- (4) Cooling fan
- (5) G sensor
- (6) Fan belt
- (7) Fan drive pulley
- (8) NE sensor
- (9) Oil drain plug
- (10) Supply pump
- (11) Engine Control Unit (ECU)

- (12) Alternator
- (13) Oil level gauge
- (14) Exhaust manifold
- (15) Flywheel
- (16) Oil pan
- (17) Starter
- (18) Water drain plug
- (19) Oil filter cartridge
- (20) Oil pressure switch

PRE-OPERATION CHECK

DAILY CHECK

To prevent trouble from occurring, it is important to know the condition of the engine well. Perform the following checks daily before starting the engine.

WARNING

To avoid personal injury or death:

- Attach shields and safeguards to the engine when operating. It is dangerous to operate with the engine's shields and safeguards removed.
 - Place the machine on a flat and open space and stop the engine before checking.
 - Keep dust or fuel away from the battery, the wiring, the muffler, and the engine in order to prevent fires.
- Check the engine and clear away any dust or fuel everyday before operating. Pay attention to the heat of the exhaust pipes and the exhaust gas so that it does not ignite trash.

Check list

Item			Ref. page
1	Check any parts that experienced trouble during previous the operation.		—
2	By walking around the machine	(1) Check for oil or cooling water (coolant) leaks.	34, 36
		(2) Check the engine oil level and make sure that the oil is not contaminated.	27
		(3) Check the fuel level.	27
		(4) Check the coolant level.	28
		(5) Check that there is no dust in the air cleaner dust cup.	32
		(6) Check for any damaged parts and loosened bolts or nuts.	—
3	By inserting the key into the starter switch	(1) Check that the meters and pilot lamps are functioning properly and that there are no stains on these parts.	—
		(2) Check that the glow lamp timer is functioning properly.	—
4	By starting the engine	(1) Check the color of exhaust fumes.	19
		(2) Check for any unusual engine noises.	18
		(3) Check the engine start-up condition.	15, 17
		(4) Check the slow-down and acceleration behavior.	18

OPERATING THE ENGINE

BREAK-IN PERIOD

During the engine break-in period, observe the following without fail:

- Change the engine oil and replace the oil filter cartridge after the first 50 hours of operation.
(See Checking oil level and adding engine oil on page 27 and Replacing oil filter cartridge on page 35.)
- When the ambient temperature is low, operate the machine only after the engine has been completely warmed up.

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

1. Starting the engine (under normal conditions)

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

WARNING

To avoid personal injury or death:

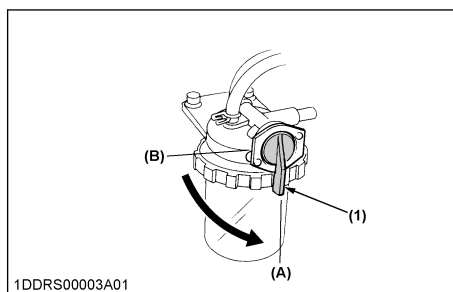
- Do not allow children to approach the machine while the engine is running.
- Place the machine in which the engine is installed, on a firm, flat and level surface.
- Do not run the engine on gradients.
- Do not run the engine in an enclosed area. Exhaust gas can cause air pollution and exhaust gas poisoning.
- Keep your hands away from rotating parts (such as the fan, the pulley, the belt, and the flywheel) during operation.
- Do not operate the machine while under the influence of alcohol or drugs.
- Do not wear loose, torn, or bulky clothing around the machine. It may catch on moving parts or controls, leading to the risk of accident. Use additional safety items (such as a hard hat, safety boots or shoes, eye and hearing protection, or gloves) as appropriate or as required.
- Do not wear radio or music headphones while operating engine.

- Check that the area around the engine is safe before starting.
- When starting the engine after performing maintenance, securely reinstall all safeguards and shields, and clear away all maintenance tools.

IMPORTANT :

- Do not use ether or any kind of starting fluid to start the engine, or a severe damage will occur.
- When starting the engine after long-term storage (more than 3 months), first set the stop lever to the "STOP" position and then activate the starter for about 10 seconds to allow oil to reach every part of the engine.

1. Set the fuel lever to the "OPEN" position.



(1) Fuel lever

(A) "OPEN"

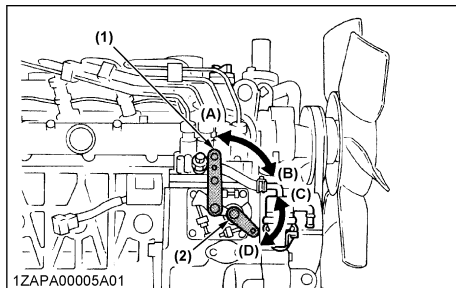
(B) "CLOSED"

2. Set the engine stop lever to the "START" position.

OPERATING THE ENGINE

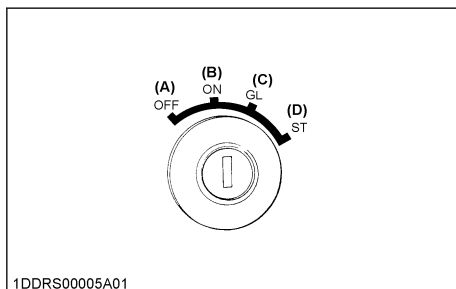
[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

- Place the speed control lever at more than half "OPERATION".



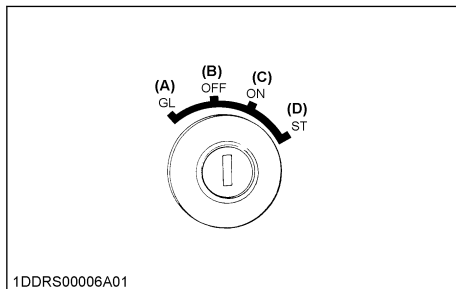
- (1) Speed control lever
(2) Engine stop lever
- (A) "IDLING"
(B) "OPERATION"
(C) "START"
(D) "STOP"

- Insert the key into the starter switch.
- Turn the key to the **[ON]** (operation) position.



- (A) **[OFF]** (Switched off)
(B) **[ON]** (Operation)
(C) **[GL]** (Preheating)
(D) **[ST]** (Starting)

KTC switch



- (A) **[GL]** (Preheating)
(B) **[OFF]** (Switched off)
(C) **[ON]** (Operation)
(D) **[ST]** (Starting)

- Turn the starter switch to the **[GL]** (preheating) position to allow the glow lamp to redden.

NOTE :

- With the lamp timer is in use, the glow lamp goes out about 5 seconds after the lamp timer is up. Refer to this for preheating. Even with the glow lamp off, the glow plug can be preheated by turning the starter switch to the **[GL]** (preheating) position.
- Turn the key to the **[ST]** (starting) position to start the engine. Release the key immediately once the engine has started.
 - Check to make sure that the oil pressure lamp and charge lamp are off. If the lamps are still on, immediately stop the engine, and determine the cause. (See CHECKS AND PRECAUTIONS DURING OPERATION on page 18.)

NOTE :

- If the oil pressure lamp is still on after the engine has started, immediately stop the engine and perform the following checks:
 - Check if there is enough engine oil.
 - Check if there is dirt in the engine oil.
 - Check if the wiring is faulty.
- Warm up the engine at medium speed without load.

IMPORTANT :

- If the glow lamp reddens too quickly or too slowly, immediately ask your Kubota dealer to check and repair it.
- If the engine does not catch or start within 10 seconds after the starter switch is set to the **[ST]** (starting) position, return the key to the **[OFF]** (switched off) position, wait for 30 seconds, and then begin the engine starting sequence again. Do not allow the starter motor to run continuously for more than 20 seconds.

2. Starting the engine (under cold conditions)

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

If the ambient temperature is below -5°C (23°F) and the engine is cold, start it in the following manner:

1. Set the fuel lever to the "OPEN" position.
2. Set the engine stop lever to the "START" position.
3. Place the speed control lever at more than half "OPERATION".
4. Insert the key into the starter switch.
5. Turn the key to the [ON] (operation) position.
6. Turn the key to the [GL] (preheating) position, and keep it there for the period of time listed in the following table.

IMPORTANT :

- The times shown in the following table are the standard preheating times for various temperatures. However, this operation is not required when the engine is already warmed up.

Ambient temperature	Preheating time	
	Ordinary heat type	With glow lamp timer
Above 10°C (50°F)	Not required	See NOTE
10°C (50°F) to -5°C (23°F)	Approximately 5 seconds	
Below -5°C (23°F)	Approximately 10 seconds	
Limit of continuous use	20 seconds	

NOTE :

- If the machine is equipped with a standard glow lamp, the glow lamp goes out about 5 seconds after the starter switch key is turned to the [GL] (preheating) position. However if necessary, keep the starter switch key at the [GL] (preheating) position for longer in according with the times listed in the left column of the previous table.

7. Turn the key to the [ST] (starting) position to start the engine.

Release the key immediately once the engine has started.

IMPORTANT :

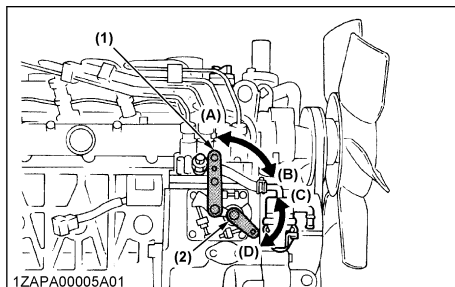
- If the engine does not catch or start within 10 seconds after the starter switch is set to the [ST] (starting) position, return the key to the [OFF] (switched off) position, wait for 30 seconds, and then begin the engine starting sequence again. Do not allow the starter motor to run continuously for more than 20 seconds.
- Always warm up the engine, not only in winter, but also in warmer seasons. Running the engine when it is not sufficiently warmed up can shorten its service life.
- When the ambient temperature is expected to be -15°C (5°F), remove the battery from the machine, and keep it indoors in a warm and safe location. Reinstall the battery in the machine just before the next operation. Do not store the battery in living areas. When working on or handling the battery, observe the safety warnings in this manual. Otherwise personal injury and property damage may occur.
(See CAUTIONS AGAINST BURNS AND BATTERY EXPLOSIONS on page 5.)

3. Stopping the engine

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

1. Return the speed control lever to low idle, and run the engine under idling conditions.
2. Set the engine stop lever to the "STOP" position.
3. Turn the starter switch to the [OFF] (switched off) position, and remove the key.

- Return the engine stop lever to the "START" position to be ready for the next start.



- (1) Speed control lever
(2) Engine stop lever
- (A) "IDLING"
(B) "OPERATION"
(C) "START"
(D) "STOP"

reinstall all safeguards and shields, and clear away all maintenance tools.

IMPORTANT :

- Do not use ether or any kind of starting fluid to start the engine, or a severe damage will occur.
- When starting the engine after long-term storage (more than 3 months), activate the starter for about 10 seconds to allow oil to reach every part of the engine.

- Set the fuel shutoff-valve to the "OPEN" position.
- Insert the key into the starter switch.
- Turn the key to the [ON] (operation) position and hold it there until the heater indicator turns off.

[D1105-K-E4]

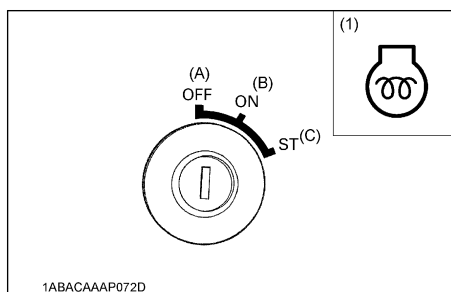
1. Starting the engine (under normal conditions)

[D1105-K-E4]

⚠ WARNING

To avoid personal injury or death:

- Do not allow children to approach the machine while the engine is running.
- Place the machine in which the engine is installed, on a firm, flat and level surface.
- Do not run the engine on gradients.
- Do not run the engine in an enclosed area. Exhaust gas can cause air pollution and exhaust gas poisoning.
- Keep your hands away from rotating parts (such as the fan, the pulley, the belt, and the flywheel) during operation.
- Do not operate the machine while under the influence of alcohol or drugs.
- Do not wear loose, torn, or bulky clothing around the machine. It may catch on moving parts or controls, leading to the risk of accident. Use additional safety items (such as a hard hat, safety boots or shoes, eye and hearing protection, or gloves) as appropriate or as required.
- Do not wear radio or music headphones while operating engine.
- Check that the area around the engine is safe before starting.
- When starting the engine after performing maintenance, securely



- (1) Heater indicator
- (A) [OFF] (Switched off)
(B) [ON] (Operation)
(C) [ST] (Starting)

- Turn the key to the [ST] (starting) position to start the engine. Release the key immediately once the engine has started.

5. Check to make sure that the oil pressure lamp and charge lamp are off. If the lamps are still on, immediately stop the engine, and determine the cause.

(See CHECKS AND PRECAUTIONS DURING OPERATION on page 18.)

NOTE :

- If the oil pressure lamp is still on after the engine has started, immediately stop the engine and perform the following checks:
 - Check if there is enough engine oil.
 - Check if there is dirt in the engine oil.
 - Check if the wiring is faulty.
 - If the oil pressure lamp stays on, and the issue is not resolved by performing these checks, contact your local Kubota dealer for service.
6. Warm up the engine at medium speed without load.

IMPORTANT :

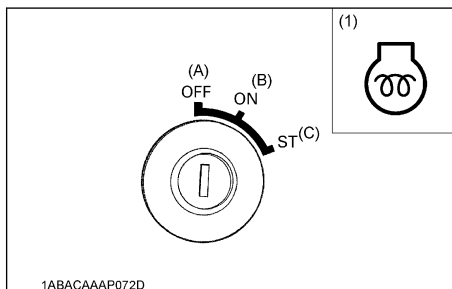
- If the engine does not catch or start within 10 seconds after the starter switch is set to the [ST] (starting) position, return the key to the [OFF] (switched off) position, wait for 30 seconds, and then begin the engine starting sequence again. Do not allow the starter motor to run continuously for more than 20 seconds.

2. Starting the engine (under cold conditions)

[D1105-K-E4]

If the ambient temperature is below -5°C (23°F) and the engine is cold, start it in the following manner:

1. Set the fuel shutoff-valve to the "OPEN" position.
2. Insert the key into the starter switch.
3. Turn the key to the [ON] (operation) position and hold it there until the heater indicator turns off.



(1) Heater indicator

- (A) [OFF] (Switched off)
(B) [ON] (Operation)
(C) [ST] (Starting)

NOTE :

- The heater indicator comes on when the key is turned to the [ON] (operation) position and the engine coolant temperature is below 0°C (-32°F).
 - The heater indicator goes off automatically when the preheat is completed.
4. Turn the key to the [ST] (starting) position to start the engine. Release the key immediately once the engine has started.

IMPORTANT :

- If the engine does not catch or start within 10 seconds after the starter switch is set to the [ST] (starting) position, return the key to the [OFF] (switched off) position, wait for 30 seconds, and then begin the engine starting sequence again. Do not allow the starter motor to run continuously for more than 20 seconds.
- Always warm up the engine, not only in winter, but also in warmer seasons. Running the engine when it is not sufficiently warmed up can shorten its service life.
- When the ambient temperature is expected to be -15°C (5°F), remove the battery from the machine, and keep it indoors in a warm and safe location. Reinstall the battery in the machine just before the next operation. Do not store the battery in living areas. When working on or handling the battery, observe the safety warnings in this manual.

Otherwise personal injury and property damage may occur.

(See **CAUTIONS AGAINST BURNS AND BATTERY EXPLOSIONS** on page 5.)

3. Stopping the engine

[D1105-K-E4]

WARNING

To avoid personal injury or death:

- To stop the engine, always use the starter switch.

When shutting off the battery power directly with the battery switch or other such methods, first stop the engine with the starter switch and then make sure that the ECU comes to a complete stop. Otherwise trouble or a malfunction may occur.

Since operation data needs to be stored in the ECU after key off, wait more than 5 minutes when shutting off the battery power directly.

1. Run the engine under idling conditions.
2. Turn the starter switch to the **[OFF]** position to stop the engine.
3. After stopping the engine, remove the key.

CHECKS AND PRECAUTIONS DURING OPERATION

With the engine running, perform the following checks to ensure that all parts are working correctly.

1. Radiator cooling water (coolant)

WARNING

To avoid personal injury or death:

- Do not remove the radiator cap until the coolant temperature is well below its boiling point. Then loosen cap slightly to the stop position, to relieve any pressure, before removing cap completely.

If the coolant temperature warning lamp lights up or if steam or coolant does not stop squirting from the radiator overflow pipe, turn off the load

and keep the engine idling (cooling down) for at least 5 minutes to let it cool down gradually. Then stop the engine and check for the cause of the problem.

(See Checking radiator cooling water (coolant) on page 36.)

2. Oil pressure lamp

The oil pressure lamp lights up to warn the operator that the engine oil pressure has dropped below the prescribed level. If it lights up during operation or if it does not turn off even after the engine has accelerated more than 1000 rpm, immediately stop the engine and check the oil level.

(See Checking oil level and adding engine oil on page 27.)

3. Charge lamp

The charge lamp lights up while the engine is running to warn the operator that the battery charge is low or that there is no alternator output. It is normal for the charge lamp to be on when the key is in the **[ON]** position and the engine is not running.

If the charge lamp is on while the engine is running, perform the following checks:

- Check for broken cables.
- Check for poor connection at the alternator terminal.
- Check if the fan belt is too loose or damaged.

4. Fuel

WARNING

To avoid personal injury or death:

- Fluid escaping from pinholes may be invisible. Do not use hands to search for suspected leaks. Use a piece of cardboard or wood instead. If injured by escaping fluid, see a medical doctor at once. This fluid can cause gangrene or a severe allergic reaction.
- Check any leaks from fuel pipes or fuel injection pipes. When checking for leaks, use eye protection, face protection, and hand protection such as gloves, along with any other appropriate personal protective equipment and clothing that are necessary.

Do not run the fuel tank level too low or completely out of fuel. You may experience improper engine running. Additionally, fuel system bleeding may be necessary if air enters the fuel system.

(See Checking fuel level and refueling on page 27.)

5. Exhaust color

- While the engine is run within the rated output range, the exhaust should remain colorless.
- If the output slightly exceeds the rated level, the exhaust may become slightly colored with the output level kept constant.
- Running the engine continuously with dark exhaust emission may lead to engine trouble.

6. Immediately stop the engine if:

- The engine suddenly slows down or accelerates.
- You suddenly hear unusual noises coming from the engine.
- Exhaust fumes suddenly become very dark.
- The oil pressure lamp or the water temperature alarm lamp lights up.
- The engine warning lamp lights up.
- The engine produces continuous large or abnormal quantities of white smoke.

REVERSED ENGINE REVOLUTION AND REMEDIES



WARNING

To avoid personal injury or death:

- **Reversed engine operation can cause the machine to reverse and run backwards. This can lead to serious trouble.**
- **Reversed engine operation can cause exhaust gas to gush out into the intake side and ignite the air cleaner, which could catch fire.**

Reversed engine revolution must be stopped immediately since engine oil circulation is cut quickly, leading to serious trouble.

1. Signs of reversed engine revolution

- Lubrication oil pressure drops sharply. The oil pressure warning light, if used, lights up.
- Since the intake and exhaust sides are reversed, the sound of the engine changes, and exhaust gas comes out of the air cleaner.
- A louder knocking sound is heard when the engine is running backwards.

2. Remedies for reversed engine revolution

1. Immediately set the engine stop lever to the "STOP" position or turn the starter switch to the [OFF] (switched off) position to stop the engine.
2. Once the engine has stopped, check the air cleaner, intake rubber tube, and other such parts, and replace them as necessary.

EMISSION CONTROL SYSTEM

- In order to keep the engine's emissions performance within the requirements applicable to the engine's category, operate, use, and maintain the engine, including the emission control system, in accordance with the instructions provided to the end users.
- Do not deliberately tamper with or misuse the engine emission control system. In particular, never deactivate the exhaust gas recirculation (EGR) system or the reagent dosing system, and always keep them properly maintained.
- When a warning indicator lights up, the engine is experiencing trouble, which may also cause problems in the emission control system. Take prompt action and rectify any incorrect operation, use, or maintenance of the emission control system in accordance with the measures described in this manual. (See TROUBLESHOOTING on page 42.)

PRECAUTIONS FOR GENERATOR EQUIPPED ENGINES



WARNING

To avoid personal injury or death:

- If the engine is kept running under no load or under a light load (load factor below 30 %), unburnt fuel or carbon builds up in the exhaust system. As a result, the engine may get damaged or cause a fire.

Run the engine under enough load at regular intervals to remove unburnt fuel and carbon deposits.

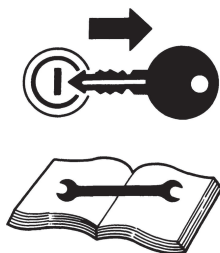
- Before performing a load operation, make sure there is nothing flammable around the engine.
- Do not put a load on the engine all at once. Increase the load gradually.
- If by any chance sparks fly out of the muffler outlet, do not unload the engine, but interrupt it immediately.

MAINTENANCE

WARNING

To avoid personal injury or death:

- Perform daily checks, periodic maintenance, refueling, or cleaning on a level surface with the engine shut off and the key removed.



1ABAAAAAP1270



1BAABADAP0010

- Before allowing other people to use your engine, explain how to operate, and have them read this manual before operation.
- Do not use gasoline to clean parts. Use a regular cleaner.
- Always use proper tools that are in good condition. Make sure that you understand how to use them before performing any service work.
- When installing parts, tighten all bolts to their specified tightening torques, making sure that none are loose.
- Do not put any tools on the battery, or battery terminals may short out. Severe burns or fire could result. Detach the battery from the engine before performing maintenance.

- Do not touch the muffler or the exhaust pipes while they are hot. Doing so could result in severe burns.

SERVICE INTERVALS

Observe the following for service and maintenance.

Interval	Item	Ref. Page		
Daily	Checking engine oil level	27		
	Checking fuel level	27		
	Checking coolant level	28		
	Checking fan belt	33		
	Cleaning evacuator valve	30		
	Cleaning dust cup equipped air cleaner (option)	30		
See NOTE.	Changing engine oil	34		
	Replacing oil filter cartridge	35		
Every 50 hours	Checking fuel pipes and clamp bands	31		
Every 100 hours	Cleaning fuel filter pot [D1005-E4, D1105-E4, D1305-E4, V1505-E4]	32		
	Cleaning air cleaner element	32	*1	
	Checking fan belt tension	33		
	Draining water separator	34		
Every 200 hours	Checking intake air line	36		
Every 200 hours or every 6 months	Checking radiator hoses and clamp bands	36		
Every 400 hours	Replacing fuel filter cartridge	36	*2	
	Cleaning water separator	37		
Every 500 hours	Removing sediment in fuel tank	37	*3	
	Cleaning radiator (inside)	37		
	Replacing fan belt	33	*3	
Every 1 year	Replacing air cleaner element	37	*4	
Every 800 hours	Checking valve clearance	37	*3	
Every 1500 hours	Checking fuel injector	37	*3	@
Every 3000 hours	Checking injection pump [D1005-E4, D1105-E4, D1305-E4, V1505-E4]	37	*3	@
Every 2 years	Replacing radiator hoses and clamp bands	36	*3	
	Replacing fuel pipes and clamp bands	31	*2 *3	
	Changing radiator coolant (LLC)	38		
	Replacing intake air line	38	*5	

*1 The air cleaner should be cleaned more often in dusty conditions than in normal conditions.

*2 When using biodiesel fuel, replace the fuel filter cartridge, fuel hoses and, clamp bands with new ones at half intervals.

*3 Consult your local Kubota dealer for this service.

*4 Replace every year or after 6 cleanings, whichever comes first.

*5 Replace only if necessary.

IMPORTANT :

- When the battery is used for less than 100 hours in a year, check its electrolyte yearly (for refillable batteries only).
- The items listed above (@ marked) are registered as emission related critical parts by Kubota in the U.S. EPA nonroad emissions regulations. As the engine owner, you are responsible for the performance of the required maintenance on the engine as listed in the previous table. Read the Warranty Statement carefully.
- Failure to perform the maintenance will cause problems that will significantly degrade the engine performance.

NOTE :

- The service intervals for the engine oil change and the oil filter cartridge replacement vary depending on the depth of the oil pan.

Engine oil change and oil filter cartridge replacement intervals

Model	Item	Oil pan depth	Initial	Interval
D1005-E4 D1105-E4 D1305-E4 V1505-E4	Engine oil	125 mm (4.9 in.) or above (110 mm D1305-E4)	50 hours	200 hours
	Oil filter cartridge	Below 101 mm (4.0 in.)*1		150 hours

*1 Oil pans below 101 mm (4.0 in.) are optional.

Model	Item	Initial	Interval
D1105-K-E4	Engine oil	50 hours	250 hours
	Oil filter cartridge		500 hours

- For engine oil grades, see **RECOMMENDED OILS AND FUELS** on page 24.
- Ambient temperature: below 35°C (95°F)

RECOMMENDED OILS AND FUELS

Engine oil

Refer to the following table for the suitable American Petroleum Institute (API) classification of engine oil according to the engine model and the fuel type used (ultra low sulfur, low sulfur, or high sulfur fuel).

Fuel type	Engine oil classification (API classification)	
	D1005-E4, D1105-E4, D1305-E4, V1505-E4	D1105-K-E4
High sulfur fuel [0.05 % (500 ppm) ≤ Sulfur content < 0.50 % (5000 ppm)]	CF (TBN≥10) (If the “CF-4, CG-4, CH-4, or CI-4” engine oil is used with a high sulfur fuel, change the engine oil at shorter intervals [approximately half].)	—
Low sulfur fuel [0.0015 % (15ppm) ≤ Sulfur content < 0.05 % (500ppm)]	CF (TBN≥10), CF-4, CG-4, CH-4, or CI-4	
Ultra low sulfur fuel [Sulfur content < 0.0015 % (15 ppm)]		CF (TBN≥10), CF-4, CG-4, CH-4, or CI-4

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

- If using a fuel with a sulfur content of 0.5 % (5000ppm) or more but less than 1.0 % (10000ppm), change the engine oil in half of the interval from the table above.

IMPORTANT :

- The engine oil used should be MIL-L-2104C or have properties of American Petroleum Institute (API) classification CF or higher.
- Change the type of engine oil according to the ambient temperature.

Above 25°C (77°F)	10W-30, 10W-40, or 15W-40
-10°C to +25°C (14°F to 77°F)	10W-30, 10W-40, or 15W-40
Below -10°C (14°F)	10W-30 or 10W-40

- When using oil different from the previous one, drain all the previous oil before adding the new engine oil.
- Use of the following oil not only prevents the engine from performing well in low temperatures until it warms up sufficiently, but may also lead to excessive engine wear or engine seizure.
 - Oil with a viscosity higher than 10W-30
 - Oil not recommended by Kubota

Fuel

- Cetane rating: The minimum recommended fuel cetane rating is 45. A cetane rating greater than 50 is preferred, especially for ambient temperatures below -20°C (-4°F) or elevations above 1500 m (5000 ft).
- The diesel fuel specification type and the sulfur content % (ppm) used must be compliant with all applicable emissions regulations for the area in which the engine is operated.
- Diesel fuels specified to EN 590 or ASTM D975 are recommended.
- No.2-D is a distillate fuel of lower volatility for engines in industrial and heavy mobile service (SAE J313 JUN87).
- When operating this engine in US EPA regulated areas, the use of ultra low sulfur fuel is mandatory. Therefore, use No.2-D S15 diesel fuel as an alternative to No.2-D, and use No.1-D S15 diesel fuel as an alternative to No.1-D for ambient temperatures below -10 °C (14°F).
- If the engine is to be operated within the European Union on diesel or non-road gas-oil, a fuel with a sulfur content no greater than 10 mg/kg (20 mg/kg at point of final distribution), a cetane

- rating no less than 45, and a fatty acid methyl ester (FAME) content no greater than 7 % volume per volume (v/v) must be used.
- When operating this engine in MEE regulated areas, use China VI (0.001 % [10ppm] sulfur maximum) or equivalent diesel fuel.

[D1105-K-E4]

- Do not use fuels that have a sulfur content greater than 0.0015 % (15 ppm).

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

- Use of diesel fuel with a sulfur content less than 0.10 % (1000 ppm) is strongly recommended.
- Do not use fuels that have a sulfur content greater than 1.0 % (10000 ppm).

List of abbreviations

ASTM	American Society of Testing and Materials
EN	European Norm
SAE	Society of Automotive Engineers
ULSD	Ultra Low Sulfur Diesel
US EPA	United States Environmental Protection Agency
MEE	Ministry of Ecology and Environment of the People's Republic of China

1. Biodiesel fuel (BDF)

When using biodiesel fuel, make sure that it meets the latest regional standards. See the following table.

Fuel type	Applicable standard		
	U.S.	Europe	China
Mineral oil diesel fuel used to make BDF	ASTM D975	EN590	GB19147
B100 fuel used to make BDF	ASTM D6751	EN14214	GB25199
The final blended fuel B5	ASTM D975	EN590	GB25199
The final blended fuel B7	—	EN590	—

Precautions in handling biodiesel fuels

- Keep the fuel tank full whenever possible to prevent water vapor from accumulating inside the fuel tank. Tighten up the fuel tank filler cap to avoid the entry of moisture.
- Routinely check the oil level before the operation. Also strictly follow the specified oil change intervals.
- Biodiesel fuels (BDF) easily deteriorate during the supply process or while in the machine due to oxygen, water, heat and other foreign substances. With this in mind, take the following precautions:
 - Do not leave biodiesel fuels in the fuel tank or a metallic drum longer than 3 months.
 - Before storing the machine for a prolonged period, change the biodiesel fuel for a conventional type of diesel fuel and operate the machine for 30 minutes or longer to clean up the fuel system.
- Bear it in mind that biodiesel fuels have the following characteristics:
 - Biodiesel fuels easily induce the growth of microorganisms and foul themselves. This may get the fuel system corroded and the fuel filter clogged.
 - In cold weather, problems such as clogging of the fuel line or fuel system, starting failure, and other unforeseen troubles may occur.
 - Biodiesel fuels easily soak up moisture, which means that they may contain higher moisture content than conventional diesel fuels.

Referring to the servicing intervals specified in the Kubota product's Operator's Manuals, be sure to maintain and clean up the fuel system, replace the fuel rubber piping with new ones and take other necessary measures. It is advisable to replace the fuel filter with a new one after half the specified replacement intervals.

(Compared with the use of mineral oil diesel fuels, the filtration performance of fuel filters gets degraded earlier than expected.)

- Palm oil-based biodiesel fuels are inferior in low-temperature fluidity to soy-based and rapeseed-based biodiesel fuels. In cold season in particular, this may clog the fuel filter.
- If biodiesel fuels are spilled on a coated surface, the coating may get damaged. Immediately wipe the spill off the surface.

DAILY CHECKS

For your own safety and to assure the long life of your machine, perform a thorough check before each operation.

1. Checking oil level and adding engine oil

WARNING

To avoid personal injury or death:

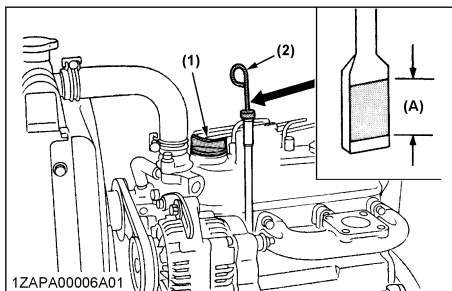
- Stop the engine before checking and changing the engine oil and the oil filter cartridge.
- Do not touch the muffler or the exhaust pipes while they are hot. Severe burns could result. Always stop the engine and allow it to cool before conducting inspections, maintenance, or cleaning.
- Contact with engine oil can damage your skin. Put on gloves before using engine oil. If you come in contact with engine oil, wash it off immediately.

NOTE :

- Place the engine on a level surface before performing inspections. If the engine is on a slope or at an angle, it is not possible to accurately measure the oil quantity.

Check the engine oil level before starting or more than 5 minutes after stopping the engine.

1. Remove the oil level gauge, wipe it clean, and reinsert it.
2. Take the oil level gauge out again, and check the oil level.



(1) Oil level gauge

(2) Oil filler plug

(A) Required oil level range
(lower end of engine oil gauge)

3. If the oil level is too low, remove the oil filler plug, and add new oil to the prescribed level.
4. After adding oil, wait for at least 5 minutes and then check the oil level again. It takes some time for the oil to come down to the oil pan.

Engine oil quantity

Model	Quantity
D1005-E4, D1105-E4	5.1 L (1.35 U.S.gals)
D1305-E4	5.7 L (1.51 U.S.gals)
V1505-E4	6.0 L (1.59 U.S.gals)
D1105-K-E4	6.2 L (1.64 U.S.gals)

Oil quantities shown are for standard oil pans.

2. Checking fuel level and refueling

Fuel is flammable and can be dangerous. Always handle fuel with care.

WARNING

To avoid personal injury or death:

- Do not mix gasoline or alcohol with diesel fuel. This mixture can cause an explosion.
- Be careful not to spill fuel during refueling. If any fuel spills, wipe it off at once, or it may cause a fire.
- Always stop the engine before refueling. Keep the engine away from the fire.
- Stop the engine before performing the following maintenance:
 - Refueling
 - Bleeding the fuel system
 - Cleaning or changing the fuel filter
 - Cleaning or changing the fuel pipes
- Do not smoke when working around the battery or when refueling.
- Check the fuel systems in a spacious and well ventilated place.
- If fuel and/or lubricant are spilled, allow the engine to cool down before refueling.
- Always keep spilled fuel and lubricant away from engine.

IMPORTANT :

- Use a strainer when filling the fuel tank. If unstrained, dirt or sand in the fuel may cause trouble in the fuel injection pump or supply pump.

- Do not run the fuel tank level too low or completely out of fuel. You may experience improper engine running. Additionally, fuel system bleeding may be necessary if air enters the fuel system.
- Fuel must be controlled carefully to prevent the additives being mixed into fuel such as fuel tank antioxidants, water remover, antifreeze and so on.

1. Check to see that the fuel level is above the lower limit of the fuel level gauge.
2. If the fuel is too low, add fuel to the upper limit. Do not overfill.

3. Checking coolant level and adding coolant

The coolant will last for one day's work if filled all the way up before operating the engine. Make it a rule to check the coolant level before every operation.

WARNING

To avoid personal injury or death:

- Do not stop the engine suddenly. Stop it after about 5 minutes of unloaded idling.
- Work only after letting the engine and the radiator cool off completely (more than 30 minutes after it has been stopped).
- Do not remove the radiator cap while the coolant is hot. Once the cap is cool to the touch, rotate it to the first stop to allow excess pressure to escape. Then remove cap completely.

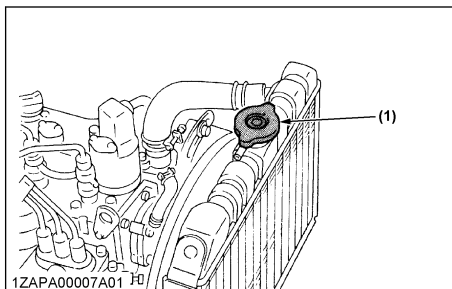
If the engine overheats, steam may gush out from the radiator or the recovery tank, and cause severe burns.

IMPORTANT :

- If the radiator cap needs to be removed, follow the safety warnings and securely retighten the cap.
- If any coolant leaks, consult your local Kubota dealer.
- Make sure that muddy water or sea water does not enter the radiator.
- Use clean, fresh water and 50 % antifreeze to fill the recovery tank.
- Do not refill the recovery tank with coolant over the [FULL] level mark.
- Close the radiator cap securely. If the cap is loose or improperly closed,

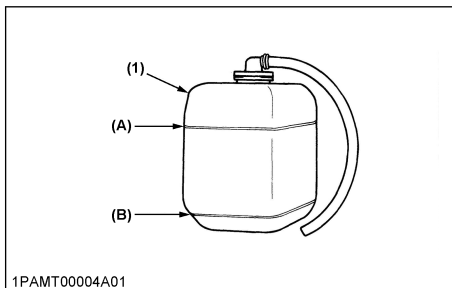
coolant may leak out and decrease quickly.

1. Allow the engine to cool down completely.
2. Remove the radiator cap and check to see that coolant reaches the supply port.



(1) Radiator pressure cap

3. If the radiator is equipped with a recovery tank, check the coolant level of the recovery tank. When it is between the [FULL] and [LOW] marks, the coolant will last for one day's work.

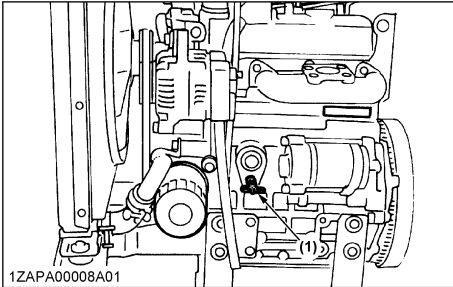


(1) Recovery tank

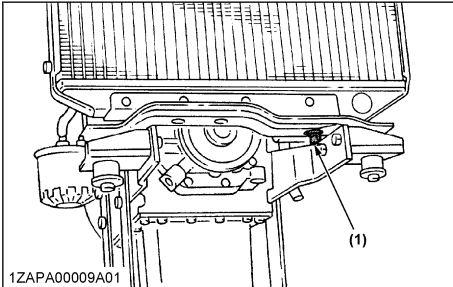
(A) [FULL]
(B) [LOW]

4. When the coolant level drops due to evaporation, add water only up to the full level.

5. Check to make sure that the 2 drain plugs are closed. One is located on the crankcase side and the other is on the lower part of the radiator.



(1) Coolant drain plug



(1) Coolant drain plug

3.1 Antifreeze

WARNING

To avoid personal injury or death:

- When handling antifreeze, wear protection such as rubber gloves (antifreeze contains poison).
- If it is swallowed, seek immediate medical help. Do not make a person throw up unless told to do so by a poison control or a health care professional. Use standard first aid and CPR for signs of shock or cardiac arrest. Call your local Poison Control Center or your local emergency number for further assistance.
- When antifreeze comes in contact with the skin or clothing, wash it off immediately.
- Do not mix different types of antifreeze. The mixture can produce chemical reaction causing harmful substances.

- Antifreeze is extremely flammable and explosive under certain conditions. Keep antifreeze away from fire and out of reach of children.
- When draining fluids from the engine, place a container underneath the engine body.
- Do not pour waste onto the ground, down a drain, or into any water source.
- Observe the relevant environmental protection regulations when disposing of antifreeze.

NOTE :

- The following data represent industry standards that necessitate minimum glycol content in the concentrated antifreeze.
- Always use a 50/50 mix of long-life coolant and clean soft water in Kubota engines.
- Contact Kubota concerning coolant for extreme conditions.
- Long-life coolant (hereafter referred to as LLC) comes in several types. Use ethylene glycol (EG) type for this engine.
- Before utilizing LLC-mixed cooling water, flush the radiator with fresh water. Repeat this procedure 2 or 3 times to clean up the radiator and the engine block from inside.
- Mixing LLC:
Premix 50 % LLC with 50 % clean soft water. When mixing, stir it up well, and then pour it into the radiator.
- The procedure for the mixing of water and antifreeze differs according to the make of the anti-freeze. Refer to SAE J1034 standard, and for more specific information also refer to SAE J814c.

Antifreeze volume	Freezing Point	Boiling Point
50 %	-37°C (-34°F)	108°C (226°F)

The specified boiling point is at 1.013 x 10⁵Pa (760 mmHg) pressure (atmospheric). A higher boiling point can be obtained by using a radiator pressure cap which permits the development of pressure within the cooling system.

- Adding LLC
 - If the coolant level in the coolant system reduces due to evaporation, top it up with water only.
 - If there is a coolant leak, add the LLC of the same manufacturer and type in the same coolant percentage.

IMPORTANT :

- Never add any long-life coolant from a different manufacturer.

Different brands may have different additive components, and the engine may fail to perform as specified.

- When the LLC is mixed, do not use any radiator cleaning agent. The LLC contains an anti-corrosive agent. If this agent is mixed with a cleaning agent, sludge may build up and adversely affect the engine parts.
- Kubota's genuine long-life coolant has a service life of 2 years. Change the coolant every 2 years.

4. Checking fan belt

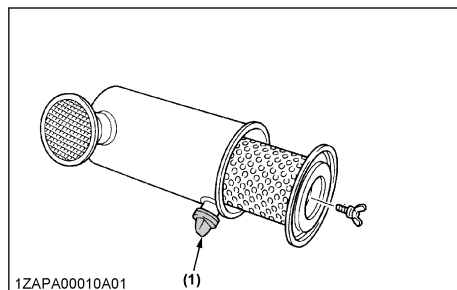
(See Adjusting fan belt tension on page 33.)

5. Cleaning evacuator valve

1. Open the evacuator valve to get rid of large particles of dust and dirt.

IMPORTANT :

- This should be done once a week under ordinary conditions, or daily when the engine is used in a dusty place.



(1) Evacuator valve

6. Cleaning dust cup equipped air cleaner (option)

1. Remove and clean out the dust cup before it becomes half full with dust.

IMPORTANT :

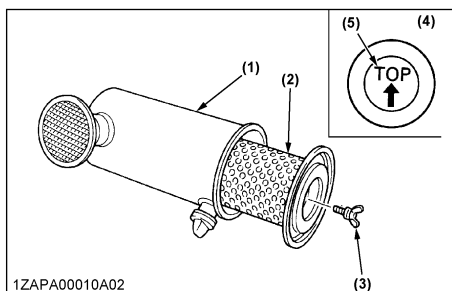
- This should be done once a week under ordinary conditions, or daily

when the engine is used in a dusty place.

2. Install the air cleaner dust cup with [TOP] mark on the rear of the cup facing upward. However, it may be installed in either direction when the cover is placed at the lower part.

IMPORTANT :

- If the dust cup is mounted incorrectly, dust and dirt do not collect in the cup. Instead, dust and dirt attach directly to the air cleaner element, which shortens its service life by a great extent.



- (1) Air cleaner body
- (2) Air cleaner element
- (3) Wing bolt
- (4) Dust cup
- (5) [TOP] mark

REGULAR CHECKS AND MAINTENANCE WORK

EVERY 50 SERVICE HOURS

1. Checking fuel pipes and clamp bands



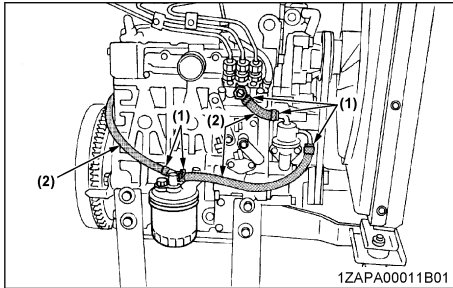
WARNING

To avoid personal injury or death:

- Check or replace the fuel pipes after stopping the engine. Broken fuel pipes can cause fires.

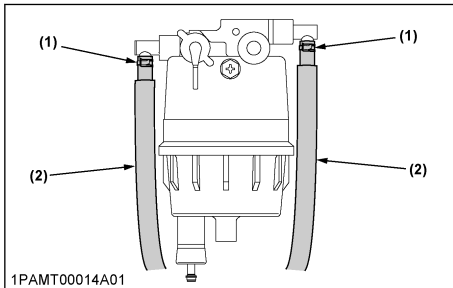
Check the fuel pipes every 50 hours of operation.

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]



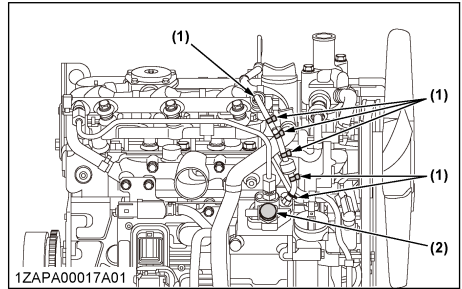
- (1) Clamp band
(2) Fuel pipe

[D1105-K-E4]



- (1) Clamp band
(2) Fuel pipe

[D1105-K-E4]



- (1) Clamp band
(2) Joint bolt

1. Check to make sure that all fuel pipes and clamp bands are tightened and not damaged.
2. Take the following measures as necessary:
 - If a clamp band is loose, apply oil to the screw of the band, and tighten the band securely.
 - If the fuel pipes, which are made of rubber, become worn out, replace them along with the clamp bands every 2 years.
 - If the fuel pipes and clamp bands are found to be worn or damaged before 2 years have passed, replace or repair them at once.
 - After replacing the pipes and bands, air bleed the fuel system. (See Bleeding fuel system on page 38.)

IMPORTANT :

- When the fuel pipes are not installed, plug them at both ends with clean cloth or paper to prevent dirt from entering. Dirt in the pipes can cause a malfunction of the fuel injection pump or supply pump.

EVERY 100 SERVICE HOURS

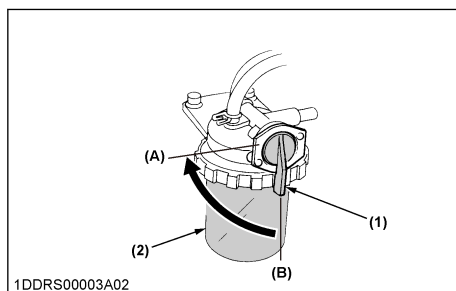
1. Cleaning fuel filter pot [D1005-E4, D1105-E4, D1305-E4, V1505-E4]

IMPORTANT :

- Dirt and dust entering the fuel line can cause a malfunction of the fuel injection pump and the injection nozzle. Wash the fuel filter cup periodically.

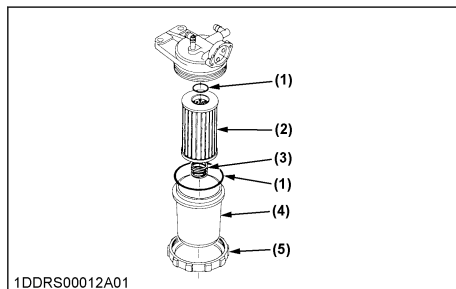
Always work in a clean place to prevent dust from entering the fuel filter or other parts.

1. Turn the fuel filter lever to the "CLOSED" position.



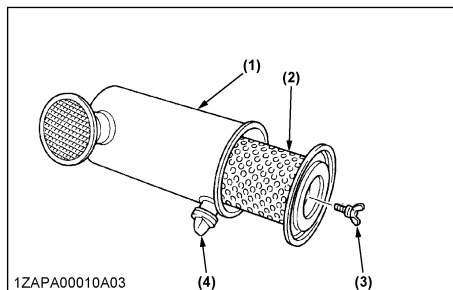
- (1) Fuel filter lever (A) "CLOSED"
(2) Fuel filter pot (B) "OPEN"

2. Remove the top cap, and rinse the inside with diesel fuel.
3. Take out the element, and rinse it with diesel fuel.
4. After cleaning, reinstall the fuel filter, making sure that no dust or dirt get into the fuel filter or other parts.
5. Air bleed the injection pump.



- (1) O ring
(2) Filter element
(3) Spring
(4) Filter bowl
(5) Screw ring

2. Cleaning air cleaner



- (1) Air cleaner body
(2) Air cleaner element
(3) Wing bolt
(4) Evacuator valve

1. Remove the wing bolt and the element.
2. Open the evacuator valve to get rid of large particles of dust and dirt.

IMPORTANT :

- This should be done once a week under ordinary conditions, or daily when the engine is used in a dusty place.

3. Wipe the inside air cleaner clean with a cloth or the like if it is dirty or wet.

IMPORTANT :

- As the air cleaner element is a dry type element, never apply oil to it.
- Avoid touching the air cleaner element except when cleaning.

4. If any dry dust has adhered to the air cleaner, blow it with compressed air from the inside while turning the element. The pressure of the compressed air must be under 205 kPa (2.1 kgf/cm², 30 psi).
5. Replace the air cleaner element with a new one if necessary.

IMPORTANT :

- The air cleaner element must be replaced every year or after 6 cleanings, whichever comes first.

6. Reassemble the air cleaner.

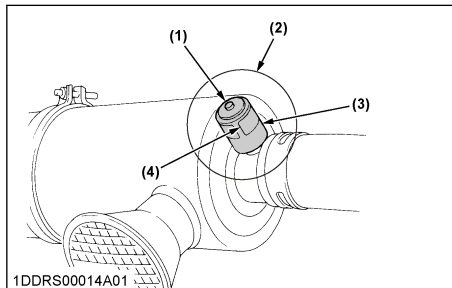
IMPORTANT :

- Make sure that the wing bolt for the element is tight enough. If it is loose, dust and dirt may be sucked in, which causes the cylinder liner and the piston ring to wear down more quickly, resulting in decreased power output.
- Do not overservice the air cleaner element. Overservicing may cause dirt to enter the engine causing premature wear. Use the dust indicator (if equipped) as a guide on when to service.

2.1 Dust indicator (option)

If the red signal on the dust indicator attached to the air cleaner is visible, the air cleaner has reached the service level.

Clean the element immediately, and reset the signal with the "RESET" button.



- (1) "RESET" button
(2) Dust indicator
(3) Service level
(4) Signal

3. Adjusting fan belt tension

WARNING

To avoid personal injury or death:

- Stop the engine and remove the key before checking the belt tension.
- Reinstall the detached safety shield after maintenance or checking.

1. Stop the engine and remove the key.
2. Apply moderate thumb pressure to the belt between the pulleys.
3. If the tension is incorrect, loosen the alternator mounting bolts, and pull the

alternator out until the belt tension reaches the proper range using a lever placed between the alternator and the engine block.

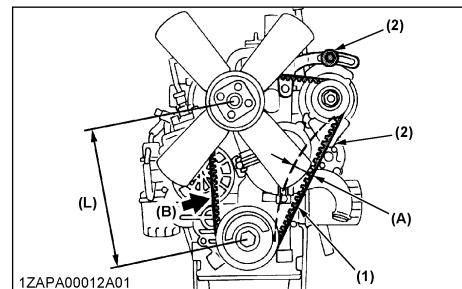
4. Replace the fan belt if it is damaged.

IMPORTANT :

- If the belt is loose or damaged, or if the cooling fan is damaged, it could result in overheating or insufficient charging. Adjust or replace belt.
- Use a sonic tension meter to measure tension. Make sure to measure properly according to the operation manual of the device being used.
- Measure immediately after installing the belt without rotating the pulleys. Measure at location (B) in the following figure with (L) as the distance between the pulleys.
- For belt specifications, contact your Kubota dealer.
- If a tension meter cannot be used, adjust based on the deflection.

The deflection must be as indicated by (A) in the following figure when pressing the middle of the belt between the pulleys.

Proper fan belt tension		
Adjustment	Replacement	(A)
200 N to 300 N	267 N to 361 N	Approximately 10 mm (0.39 in.) Under load of 10 kgf (22.1 lbs)



- (1) Fan belt
(2) Bolt and nut
(A) Deflection
(B) Measuring point
(L) Distance between the pulleys

4. Draining water separator

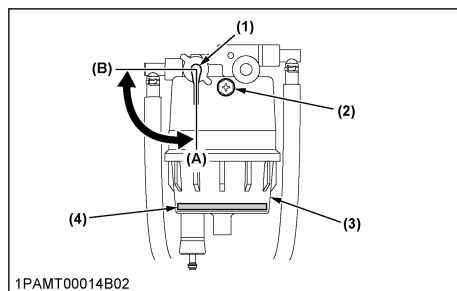
Water and dirt contained in fuel will settle in the water separator. When deposits of such foreign substances build up, drain and clean the water separator as follows.

Draining

1. Set the fuel shutoff-valve to the "CLOSED" position.
2. Loosen the top screw first and then loosen the bottom drain valve to let the foreign substances out of the separator.
3. Air-bleed the fuel system.
(See Bleeding fuel system on page 38.)

Cleaning

1. Set the fuel shutoff-valve to the "CLOSED" position.
2. Loosen and remove the cup properly, and clean its inside with diesel fuel.
3. Tighten up the cup properly.



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- | | |
|------------------------|--------------|
| (1) Fuel shutoff-valve | (A) "OPEN" |
| (2) Screw | (B) "CLOSED" |
| (3) Cup | |
| (4) Float | |

IMPORTANT :

- As water is collected, the red float goes up.
Drain the water separator before the float reaches the top of the cup.
- In reattaching the water separator, be careful to keep off dust and dirt.
- Always air-bleed the fuel system before restarting the engine.

EVERY 200 SERVICE HOURS

1. Changing engine oil

WARNING

To avoid personal injury or death:

- Stop the engine before draining the engine oil.
- When draining the engine oil, place a container underneath the engine, and dispose of the oil according to local regulations.
- Do not drain the oil immediately after running the engine. Allow engine to cool down sufficiently.

Change oil after the initial 50 hours of operation and then at the intervals listed in the following table thereafter.

Engine oil change intervals

Model	Oil pan depth	Initial	Interval
D1005-E4 D1105-E4 D1305-E4 V1505-E4	125 mm (4.9 in.) or above (110 mm D1305-E4)	50 hours	200 hours
	Below 101 mm (4.0 in.) ^{*1}		150 hours

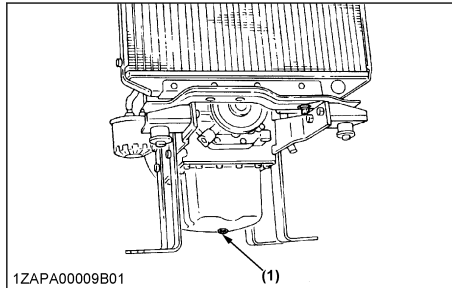
^{*1} Oil pans below 101 mm (4.0 in.) are optional.

Model	Initial	Interval
D1105-K-E4	50 hours	250 hours

- For engine oil grades, see RECOMMENDED OILS AND FUELS on page 24.
- Ambient temperature: below 35°C (95°F)

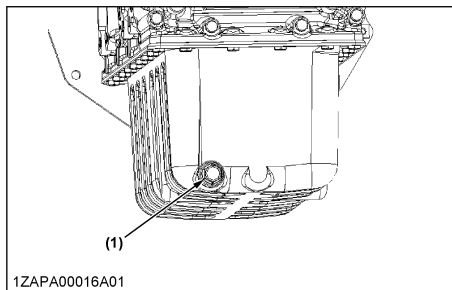
1. Remove the drain plug at the bottom of the engine, and drain out all of the old oil.

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]



(1) Oil drain plug

[D1105-K-E4]



(1) Oil drain plug

2. Add new engine oil up to the upper limit of the oil level gauge.

2. Replacing oil filter cartridge

WARNING

To avoid personal injury or death:

- Stop the engine before changing the oil filter cartridge.
- Allow the engine to cool down sufficiently. Oil can be hot and cause burns.

Replace the oil filter cartridge at the intervals listed in the following table.

Engine oil filter cartridge replacement intervals

Model	Oil pan depth	Initial	Interval
D1005-E4 D1105-E4 D1305-E4 V1505-E4	125 mm (4.9 in.) or above (110 mm D1305-E4)	50 hours	200 hours
	Below 101 mm (4.0 in.)*1		150 hours

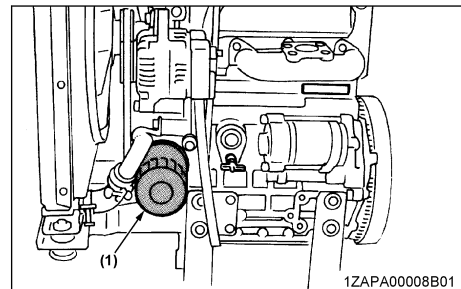
*1 Oil pans below 101 mm (4.0 in.) are optional.

Model	Initial	Interval
D1105-K-E4	50 hours	500 hours

1. Remove the old oil filter cartridge with a filter wrench.
2. Apply a film of oil to the gasket for the new cartridge.
3. Screw in the cartridge by hand. When the gasket contacts the seal surface, tighten the cartridge sufficiently by hand.

IMPORTANT :

- Do not use the filter wrench to tighten the cartridge. Using a wrench will cause the cartridge to be overtightened.



(1) Oil filter cartridge

4. Run the engine for a while and check for oil leaks through the seal.
5. Check the engine oil level, and add oil if necessary.
After the new cartridge has been installed, the engine oil level normally decreases slightly.

NOTE :

- Completely wipe off any oil that sticks to the machine.

3. Checking intake air line

Visually check for cracks, gas leaks, and anything else unusual.

Check that the lock bolts and nuts are tight enough.

EVERY 200 SERVICE HOURS OR EVERY 6 MONTHS

1. Checking radiator hoses and clamp bands

WARNING

To avoid personal injury or death:

- Check the radiator hoses and clamp bands periodically. If any one of the radiator hoses is damaged, or if coolant leaks, overheating or severe burns can occur.

Check the radiator hoses and clamp bands every 200 hours of operation or every 6 months, whichever comes first.

1. Check the hoses and clamp bands to make sure that they are properly connected.
2. If any one of the hose clamps is loose, or if any coolant is leaking, tighten the hose clamp securely.
3. If any of the radiator hoses are swollen, hardened or cracked, replace them with new hoses and tighten the hose clamps securely.

IMPORTANT :

- Replace the hoses and the hose clamps at least every 2 years. If they are found to be swollen, hardened, or cracked, replace them immediately.

1.1 Overheating countermeasures

When the coolant temperature gets close to or exceeds the boiling point, this is called "overheating".

With the engine running check the parts to make sure that they are working correctly.

If anything is unusual, refer to the relevant sections of this operator's manual.

(See MAINTENANCE on page 21 and REGULAR CHECKS AND MAINTENANCE WORK on page 31.)

1.1.1 Checking radiator cooling water (coolant)

WARNING

To avoid personal injury or death:

- Do not remove the radiator cap until the coolant temperature is well below its boiling point. Then loosen cap slightly to the stop position, to relieve any pressure, before removing cap completely.

If the coolant temperature warning lamp lights up or if steam or coolant does not stop squirting from the radiator overflow pipe, take the following measures:

1. Remove the engine load.
2. Keep the engine idling (cooling down) for at least 5 minutes to let it cool down gradually.
3. Stop the engine.
4. Check the radiator as follows, and perform any necessary maintenance:
 - a. Check to see if the coolant runs short or if there are any coolant leaks.
 - b. Check to see if there is any foreign matter around the cooling air inlet or the outlet.
 - c. Check to see if there is any dirt or dust between the radiator fins and the tubes.
 - d. Check to see if the fan belt is too loose.
 - e. Check to see if the radiator water pipes are clogged.

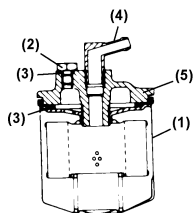
EVERY 400 SERVICE HOURS

1. Replacing fuel filter cartridge

IMPORTANT :

- Replace the fuel filter cartridge periodically to prevent wear of the fuel injection pump or supply pump plunger or the injection nozzle due to dirt in the fuel.

Replace the fuel filter cartridge with a new one every 400 operating hours.



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- (1) Fuel filter cartridge
- (2) Air vent plug
- (3) O ring
- (4) Pipe joint
- (5) Cover

1. Remove the existing fuel filter cartridge.
2. Apply a thin layer of fuel over the gasket.
3. Tighten the new cartridge into position by hand-tightening only.
4. Vent the air.

2. Cleaning water separator

(See Draining water separator on page 34.)

EVERY 500 SERVICE HOURS

1. Removing sediment in fuel tank

Consult your local Kubota dealer for this service.

2. Cleaning radiator core (outside)

1. Check for dust between the fins and tubes.
2. Wash away any dust with running water.

IMPORTANT :

- Do not clean the radiator with hard tools such as spatulas or screwdrivers, as they may damage the fins or tubes, causing coolant leaks or decreased cooling performance.

3. Cleaning radiator (inside)

- Clean up the coolant line inside in the following cases:
 - When cleaning the radiator as per the service intervals list (See SERVICE INTERVALS on page 22.)
 - When changing the coolant

- Use a radiator cleaning agent. This helps wash away scale deposits.

4. Replacing fan belt

Consult your local Kubota dealer for this service.

EVERY 1 YEAR

1. Replacing air cleaner element

- Replace every year or after 6 cleanings, whichever comes first.
 - Consult your local Kubota dealer for this service.
- (See Cleaning air cleaner on page 32.)

EVERY 800 SERVICE HOURS

1. Checking valve clearance

Consult your local Kubota dealer for this service.

EVERY 1500 SERVICE HOURS

1. Checking fuel injector

Consult your local Kubota dealer for this service.

EVERY 3000 SERVICE HOURS

1. Checking injection pump

Consult your local Kubota dealer for this service.

EVERY 2 YEARS

1. Replacing radiator hoses and clamp bands

Consult your local Kubota dealer for this service.
(See Checking radiator hoses and clamp bands on page 36.)

2. Replacing fuel pipes and clamp bands

Consult your local Kubota dealer for this service.

3. Changing radiator coolant (LLC)



WARNING

To avoid personal injury or death:

- Do not stop the engine suddenly. Stop it after about 5 minutes of unloaded idling.
- Work only after letting the engine and the radiator cool off completely (more than 30 minutes after it has been stopped).
- Do not remove the radiator cap while the coolant is hot. Once the cap is cool to the touch, rotate it to the first stop to allow excess pressure to escape. Then remove cap completely.
If the engine overheats, steam may gush out from the radiator or the recovery tank, and cause severe burns.

IMPORTANT :

- If the radiator cap needs to be removed, follow the safety warnings and securely retighten the cap.
- If any coolant leaks, consult your local Kubota dealer.
- Make sure that muddy water or sea water does not enter the radiator.
- Use clean, fresh water and 50 % antifreeze to fill the recovery tank.
- Do not refill the recovery tank with coolant over the [FULL] level mark.
- Close the radiator cap securely. If the cap is loose or improperly closed, coolant may leak out and decrease quickly.

1. Open both drain plugs and the radiator cap at the same time.
It is not possible to completely drain the coolant with the radiator cap closed.
2. Remove the overflow pipe of the radiator pressure cap to drain the recovery tank.
3. Close the coolant drain plugs.
4. Fill the radiator with coolant as prescribed in the following tables.

Coolant (radiator cleaner and antifreeze)

Season	Coolant
All seasons	Pure water and antifreeze (See Antifreeze on page 29.)

Coolant volume

Model	Quantity
D1005-E4, D1105-E4, D1305-E4, D1105-K-E4	3.1 L (0.82 U.S.gals)
V1505-E4	4.0 L (1.06 U.S.gals)

NOTE :

- Coolant quantities shown are for standard radiators.

5. Retighten the radiator cap.

NOTE :

- An improperly tightened radiator cap or a gap between the cap and the seat quickens the loss of coolant.

3.1 Remedies for quick decrease of coolant

- Check for any dust or dirt between the radiator fins and tube. If any dust or dirt is found, remove it from the fins and tube.
- Check the tightness of the fan belt. If it is loose, tighten it securely.
- Check for internal blockage in the radiator hose. If any scale has formed in the radiator hose, clean it with scale inhibitor or the equivalent.

4. Replacing intake air line

Consult your local Kubota dealer for this service.

SERVICE AS REQUIRED

1. Bleeding fuel system



WARNING

To avoid personal injury or death:

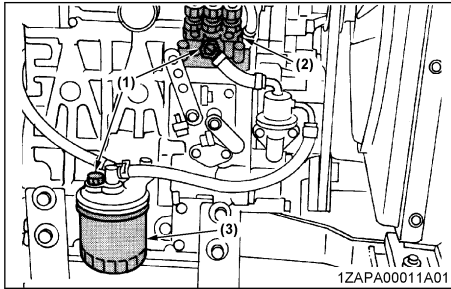
- Do not bleed a hot engine as this could cause fuel to spill onto a hot exhaust manifold creating a danger of fire.

Bleeding of the fuel system is required at the following times:

- After the fuel filter and pipes have been detached and refitted
- After the fuel tank has become empty
- Before the engine is to be used after a long storage

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]**Procedure A [gravity feed fuel tanks only]**

1. Fill the fuel tank to the fullest extent.
2. Open the fuel filter lever.
3. Loosen air vent plug of the fuel filter a few turns.
4. Screw back the plug when bubbles do not come up any more.
5. Open the air vent plug on top of the fuel injection pump.
6. Retighten the plug when bubbles do not come up any more.

Gravity feed system

- 1ZAPA00011A01
- (1) Air vent plug
 - (2) Injection pump
 - (3) Fuel filter

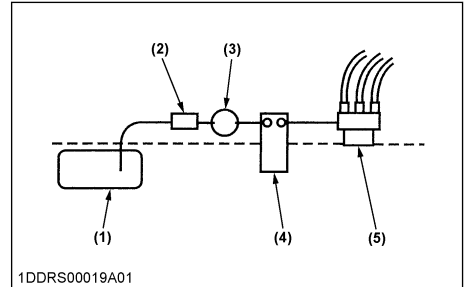
Procedure B [fuel tanks lower than injection pump]

1. Pressurize the fuel system as follows:
 - For fuel tanks that are lower than the injection pump, the fuel system must be pressurized by the fuel system electric fuel pump.
 - If an electric fuel pump is not used, you must manually actuate the pump by lever bleed.
 - The primary fuel filter must be on the pressure side of the pump if the fuel tank is lower than the injection pump.
2. Loosen air vent plug of the fuel filter a few turns.
3. Screw back the plug when bubbles do not come up any more.
4. Open the air vent plug on top of the fuel injection pump.
5. Retighten the plug when bubbles do not come up any more.

IMPORTANT :

- Keep the air vent plug of the fuel injection pump tightened when not

bleeding. Otherwise it may cause the engine to stop.

Tank below injection pump system

1DDRS00019A01

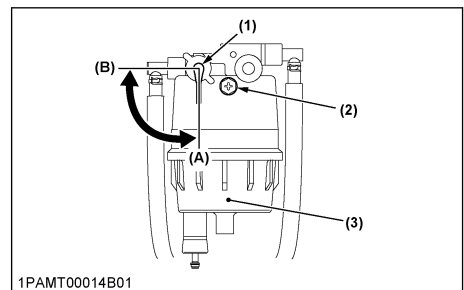
- (1) Fuel tank below injection pump
- (2) Pre-filter
- (3) Electric or mechanical pump
- (4) Main filter
- (5) Injection pump

[D1105-K-E4]

1. Loosen the screw on the water separator a few turns.
2. Retighten the screw when bubbles do not come up any more.
3. Loosen the screw on top of the supply pump.
4. Retighten the screw when bubbles do not come up any more.

IMPORTANT :

- Do not change the direction of the joint port when tightening the joint bolt.



1PAMT00014B01

- (1) Fuel shut-off-valve
- (2) Screw
- (3) Cup
- (A) "OPEN"
- (B) "CLOSED"

TRANSPORTING AND STORING THE ENGINE

TRANSPORTING THE ENGINE

WARNING

To avoid personal injury or death:

- Fix the engine securely so that it does not fall during transport.
- Do not stand near or under the engine while carrying it.
- The engine is heavy. When handling the engine, be very alert so that your hands and body do not get crushed.
- Use a carrier such as a crane to transport the engine, otherwise you may hurt your back or injure yourself. Support the engine securely with a rope so that it does not fall during transport.
- When lifting the engine, securely attach the hooks to the metal fittings on the engine. Use hooks and fittings strong enough to support the engine.

STORING THE ENGINE

WARNING

To avoid personal injury or death:

- Do not clean the machine with engine running.
- To avoid the danger of exhaust fume poisoning, do not operate the engine in a closed building without proper ventilation.
- After operation, allow the engine to cool off sufficiently before storing.

Before storing the engine for more than a few months, perform the following procedure:

1. Remove any dirt on the machine.
2. Drain the coolant from the radiator by opening the plug at the bottom of the radiator, removing the pressure cap, and allowing the coolant to drain out completely. Leave the plug open, and hang a "No coolant" on the pressure cap.

IMPORTANT :

- Since coolant may freeze when the temperature drops below 0°C (32°F), it is very important that no coolant is left in the machine.

NOTE :

- When using antifreeze, it is not necessary to drain the coolant.
3. Remove dirty engine oil, fill with new oil, and run the engine for about 5 minutes to let the oil penetrate to all the parts.
 4. Check all the bolts and nuts, and tighten if necessary.
 5. Remove the battery from the engine, adjust the electrolyte level, and recharge it. Store the battery in a dry and dark place.
 6. Store the engine in a flat place and remove the key from engine.

WARNING

To avoid personal injury or death:

- Do not store the engine near any flammable materials such as dry grass or straw.
- Before covering the engine for storage, let engine and muffler cool off completely.

IMPORTANT :

- When the engine is not used for a long period of time, run it for about 5 minutes under no load every 2 to 3 months to keep it free from rust. If the engine is not run periodically, moisture in the air may cause condensation to form on the sliding parts of the engine, resulting in rust.
- If you forget to run the engine for longer than 5 to 6 months, apply sufficient engine oil to the valve guides and valve stem seals, and make sure the valves work smoothly before starting the engine.
- Before operating the engine after storage, check the wiring and the pipes and repair any damage. Also, clean out

any flammable materials that may have been carried in by mice.

TROUBLESHOOTING

If the engine does not function properly, use the following chart to identify and correct the cause.

If the cause of trouble cannot be found, contact your Kubota dealer.

TROUBLESHOOTING CHART [D1005-E4, D1105-E4, D1305-E4, V1505-E4]

Trouble	Cause	Countermeasure
Starting difficulties	The fuel is thick and doesn't flow.	- Check the fuel tank and fuel filter. - Remove water, dirt, and other impurities. - As all fuel will be filtered by the filter, if there is any water or other foreign matter on the filter, clean the filter with kerosene.
	Air or water mixed in the fuel system	- If air is in the fuel system or injection lines, the fuel pump will not work properly. - To attain proper fuel injection pressure, check carefully to make sure that parts such as the fuel line coupling and cap nut are not loose. - Loosen the screw on the water separator and the screw on top of the fuel injection pump, to eliminate all the air in the fuel system.
	The engine oil becomes thick in cold weather and the engine cranks slow down.	Change grade of oil according to the weather (temperature). Use 10W-30 in winter season.
	The battery is discharged and the engine will not crank.	- Charge the battery. - In winter, always remove the battery from the machine, fully charge it, and keep it indoors. Install it in the machine only at time of use.
Insufficient engine power	Low fuel level	Check the fuel system.
	The moving parts are overheating.	- Check lubricating oil system. - Check to see if lubricating oil filter is working properly. - Filter element deposited with impurities would cause poor lubrication. Change element.
	Clogged air cleaner	Clean the air cleaner element every 100 hours of operation.
	Injection pump wear	Do not use poor quality fuel as it will cause wear of the pump. Only use No. 2-D diesel fuel. (See RECOMMENDED OILS AND FUELS on page 24.)
The engine suddenly stops.	Lack of fuel	- Refuel - Check the fuel system. (Bleed the fuel system if necessary.)
	Bad nozzle	If necessary, replace with a new nozzle.
	Moving parts are overheated due to shortage of lubrication oil or improper lubrication.	- Check amount of engine oil with oil level gauge. - Check lubricating oil system. - The oil filter cartridge should be replaced once for every 2 oil changes. - Check to see if the engine bearing clearances are within factory specs.
Abnormal exhaust gas color	Poor quality fuel	Use high quality fuel. Use No. 2-D diesel fuel only.
	Bad nozzle	If necessary, replace with a new nozzle.

(Continued)

Trouble	Cause	Countermeasure
The engine over- heats.	Lack of engine oil	• Check the oil level and refill if necessary.
	The fan belt is broken or elongated	• Change belt or adjust belt tension.
	Lack of coolant	• Refill coolant.
	Excessive concentration of antifreeze	• Add only water or change to coolant with the specified mixing ratio.
	The radiator net or the radiator fin is clogged with dust.	• Clean net or fin carefully.
	The inside of the radiator or the coolant flow route is corroded.	• Clean or replace radiator and parts.
	The cooling fan, the radiator, or the radiator cap is damaged.	• Replace damaged parts.
	The thermostat is damaged.	• Check the thermostat and replace if necessary.
	The temperature gauge or the sensor is damaged.	• Check the temperature with the thermometer and replace if necessary.
	Overload running	• Reduce the load.
	The head gasket is damaged or coolant is leaking.	• Replace the parts.
	The fuel used is unsuitable.	• Use the specified fuel.

NOTE :

- If the cause of trouble can not be found, contact your Kubota dealer.

Immediately stop the engine in the following situations:

Trouble	Countermeasure
The color of the exhaust suddenly turns dark.	• Check the fuel injection system, especially the fuel injection nozzle.
The bearing parts have overheated.	• Check the lubricating system.
The oil lamp lights up during operation.	• Check the lubricating system. • Check the function of the relieve valve in the lubricating system. • Check the pressure switch. • Check the filter base gasket.

TROUBLESHOOTING CHART [D1105-K-E4]

Trouble	Cause	Countermeasure
Starting difficulties	The fuel is thick and doesn't flow.	- Check the fuel tank and fuel filter. - Remove water, dirt, and other impurities. - As all fuel will be filtered by the filter, if there is any water or other foreign matter on the filter, clean the filter with kerosene.
	Air or water mixed in the fuel system	- If air is in the fuel system or injection lines, the fuel pump will not work properly. - To attain proper fuel injection pressure, check carefully to make sure that parts such as the fuel line coupling and cap nut are not loose. - Loosen the screw on the water separator and the screw on top of the supply pump, to eliminate all the air in the fuel system.
	The engine oil becomes thick in cold weather and the engine cranks slow down.	Change grade of oil according to the weather (temperature). Use 10W-30 in winter season.
	The battery is discharged and the engine will not crank.	- Charge the battery. - In winter, always remove the battery from the machine, fully charge it, and keep it indoors. Install it in the machine only at time of use.
Insufficient engine power	Low fuel level	Check the fuel system.
	The moving parts are overheating.	Consult your local Kubota dealer.
	Clogged air cleaner	Clean the air cleaner element every 100 hours of operation.
	Engine output is limited because of trouble.	Check the engine warning lamp. (If trouble occurs, it might be because the ECU is in output limiting mode.)
The engine suddenly stops.	Lack of fuel	- Refuel - Check the fuel system. (Bleed the fuel system if necessary.)
	Overheating of moving parts	Consult your local Kubota dealer.
	Air cleaner is dirty	Clean the air cleaner element.
	Forced stop due to trouble	Check the engine warning lamp. (If serious trouble occurs, it means that the ECU might have forced the engine to stop.)
Abnormal exhaust gas color	Poor quality fuel	Use high quality fuel. Use No. 2-D diesel fuel only.
The engine overheats.	Lack of engine oil	Check the oil level and refill if necessary
	The fan belt is broken or elongated	Change belt or adjust belt tension.
	Lack of coolant	Refill coolant.
	Excessive concentration of antifreeze	Add only water or change to coolant with the specified mixing ratio.
	The radiator net or the radiator fin is clogged with dust.	Clean net or fin carefully.
	The inside of the radiator or the coolant flow route is corroded.	Clean or replace radiator and parts.
	The cooling fan, the radiator, or the radiator cap is damaged.	Replace damaged parts.
	The thermostat is damaged.	Check the thermostat and replace if necessary.
	The temperature gauge or the sensor is damaged.	Check the temperature with the thermometer and replace if necessary.

(Continued)

Trouble	Cause	Countermeasure
The engine overheats.	Overload running	• Reduce the load.
	The head gasket is damaged or coolant is leaking.	• Replace the parts.
	The fuel used is unsuitable.	• Use the specified fuel.

Immediately stop the engine in the following situations:

Trouble	Countermeasure
The color of the exhaust suddenly turns dark.	• Consult your local Kubota dealer.
The bearing parts have overheated.	• Check the lubricating system.
The oil lamp lights up during operation.	• Check the lubricating system. • Check the function of the relieve valve in the lubricating system. • Check the pressure switch. • Check the filter base gasket.
Engine revolution suddenly decreases or increases.	• Check the adjustments and the fuel system.
Unusual sounds can be heard coming from the engine.	• Check all of the moving parts carefully.
The engine warning lamp lights up.	• Consult your local Kubota dealer.

SPECIFICATIONS

Model		D1005-E4	D1105-E4
Type		Vertical, water-cooled, 4-cycle diesel engine	
Number of cylinders		3	
Bore and stroke	mm (in.)	76 x 73.6 (2.99 x 2.90)	78 x 78.4 (3.07 x 3.09)
Total displacement	cm ³ (in. ³)	1001 (61.08)	1123 (68.53)
Combustion chamber		Spherical Type (E-TVCS)	
SAE NET Intermittent H.P. (SAEJ1349)	kW/rpm (HP/rpm)	17.7/3200 (23.7/3200)	17.8/3000 (23.9/3000)
SAE NET Continuous H.P. (SAEJ1349)	kW/rpm (HP/rpm)	15.4/3200 (20.6/3200)	15.4/3000 (20.7/3000)
Maximum bare speed	rpm	3420	3220
Minimum bare idling speed	rpm	1300	900
Order of firing		1-2-3	
Direction of rotation		Counter-clockwise (viewed from flywheel side)	
Injection pump		Bosch MD Type Mini Pump	
Injection pressure		13.73 Mpa (140 kgf/cm ² , 1991 psi)	
Injection timing (BTDC)		20°	18°
Compression ratio		24 : 1	
Fuel		Diesel Fuel No.2-D (ASTM D975)	
Lubricant (API classification)		CF or above	
Dimensions (length x width x height)	mm (in.)	497.8 x 396.0 x 602.0 (19.60 x 15.59 x 23.7)	
Dry weight (BB Spec.)	kg (lbs.)	93 (205.0)	
Starting system		Cell starter (with glow plug)	
Starting motor		12 V, 1.2 kW	
Charging generator		12 V, 480 W	
Recommended battery capacity		12 V, 65 AH, equivalent	
EU engine family name		HKBXL01.5BCB	
CO ₂ Emission data Non-road steady-state test cycle	g/kWh	1018.0	

NOTE :

- Specifications are subject to change without notice.
- The battery capacity is indicated in 5-hour ratio.
- This CO₂ measurement results from testing a(n) (parent) engine representative of the engine type (engine family) over a fixed test cycle under laboratory conditions, and does not imply or express any guarantee of the performance of a particular engine.

Model		D1305-E4	V1505-E4
Type		Vertical, water-cooled, 4-cycle diesel engine	
Number of cylinders		3	4
Bore and stroke	mm (in.)	78 x 88 (3.07 x 3.46)	78 x 78.4 (3.07 x 3.09)
Total displacement	cm ³ (in ³)	1261 (76.95)	1498 (91.41)
Combustion chamber		Spherical Type (E-TVCS)	
SAE NET Intermittent H.P. (SAEJ1349)	kW/rpm (HP/rpm)	17.9/2600 (24/2600)	17.7/2300 (23.7/2300)
SAE NET Continuous H.P. (SAEJ1349)	kW/rpm (HP/rpm)	15.5/2600 (20.8/2600)	15.4/2300 (20.6/2300)
Maximum bare speed	rpm	2820	2520
Minimum bare idling speed	rpm	1100	1150
Order of firing		1-2-3	1-3-4-2
Direction of rotation		Counter-clockwise (viewed from flywheel side)	
Injection pump		Bosch MD Type Mini Pump	
Injection pressure		13.73 Mpa (140 kgf/cm ² , 1991 psi)	
Injection timing (BTDC)		16°	14°
Compression ratio		24 : 1	
Fuel		Diesel Fuel No.2-D (ASTM D975)	
Lubricant (API classification)		CF or above	
Dimensions (length x width x height)	mm (in.)	497.6 x 396.0 x 590.1 (19.59 x 15.59 x 23.2)	591.3 x 396.0 x 607.0 (23.28 x 15.59 x 23.90)
Dry weight (BB Spec.)	kg (lbs.)	95 (209)	110 (242.5)
Starting system		Cell starter (with glow plug)	
Starting motor		12 V, 1.2 kW	
Charging generator		12 V, 480 W	
Recommended battery capacity		12 V, 65 AH, equivalent	12 V, 75 AH, equivalent
EU engine family name		HKBXL01.5BCB	
CO ₂ Emission data Non-road steady-state test cycle	g/kWh	1018.0	

NOTE :

- Specifications are subject to change without notice.
- The battery capacity is indicated in 5-hour ratio.
- This CO₂ measurement results from testing a(n) (parent) engine representative of the engine type (engine family) over a fixed test cycle under laboratory conditions, and does not imply or express any guarantee of the performance of a particular engine.

Model		D1105-K-E4
Type		Vertical, water-cooled, 4-cycle diesel engine
Number of cylinders		3
Bore and stroke	mm (in.)	78 x 78.4 (3.07 x 3.09)
Total displacement	cm ³ (in ³)	1123 (68.53)
Combustion type		TVCR
SAE NET Intermittent H.P. (SAEJ1349)	kW/rpm (HP/rpm)	17.8/3000 (23.9/3000)
SAE NET Continuous H.P. (SAEJ1349)	kW/rpm (HP/rpm)	15.4/3000 (20.7/3000)
Maximum bare speed	rpm	3200
Minimum bare idling speed	rpm	850 to 950
Order of firing		1-2-3
Direction of rotation		Counter-clockwise (viewed from flywheel side)
Compression ratio		23 : 1
Fuel		Diesel Fuel No.2-D (ASTM D975)
Lubricant (API classification)		CF or above
Dimensions (length x width x height)	mm (in.)	497.8 x 396.0 x 595.0 (19.60 x 15.59 x 23.4)
Dry weight (BB Spec.)	kg (lbs.)	94 (207.2)
Starting system		Cell starter (with glow plug)
Starting motor		12 V, 1.2 kW
Charging generator		12 V, 480 W
Recommended battery capacity		12 V, 55AH, equivalent
EU engine family name		RKBXL01.1BFB
CO2 Emission data Non-road steady-state test cycle	g/kWh	1063.7

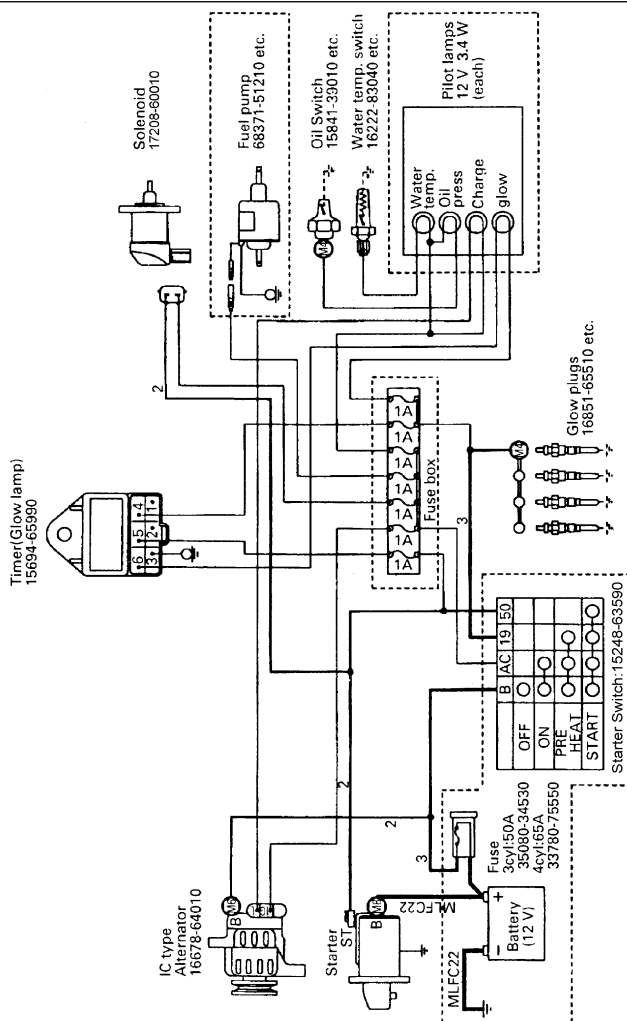
NOTE :

- Specifications are subject to change without notice.
- The battery capacity is indicated in 5-hour ratio.
- This CO2 measurement results from testing a(n) (parent) engine representative of the engine type (engine family) over a fixed test cycle under laboratory conditions, and does not imply or express any guarantee of the performance of a particular engine.

WIRING DIAGRAMS

[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

EU standard (Energize to run)



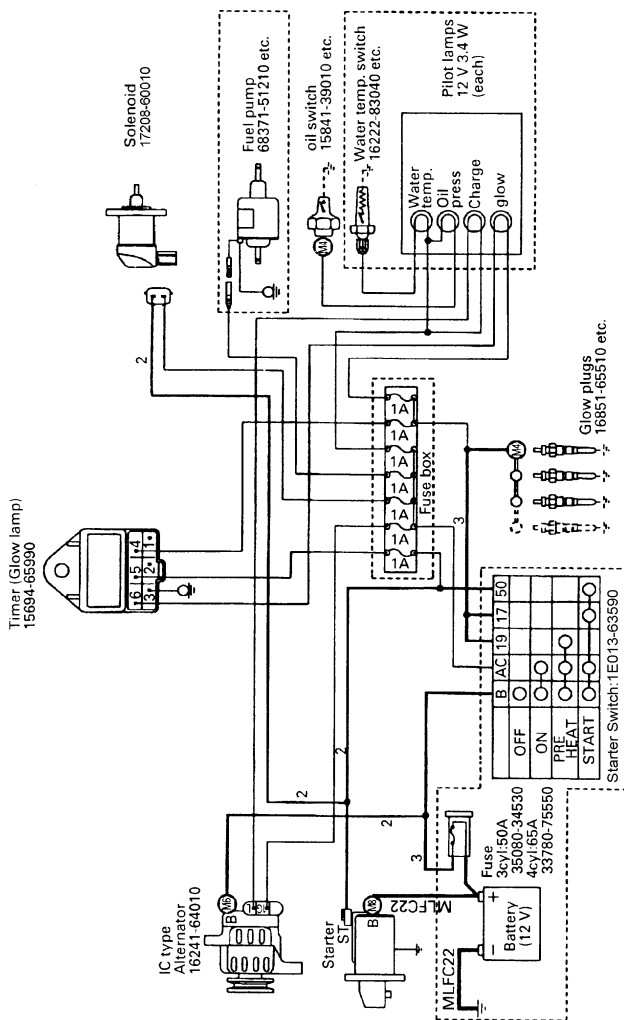
★ The parts boxed in [] are reference, NOT equipped for standard engine spec.

★ Non marked wire dia. is 0.8~1.25 mm².

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[D1005-E4, D1105-E4, D1305-E4, V1505-E4]

KEA/SAE standard (Energize to run)



★ The parts boxed in [] are reference, NOT equipped for standard engine spec.

★ Non marked wire dia. is 0.8~1.25 mm².

1ZAPA00014A01enUS

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